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Examining the role of CEO's authentic communication on corporate reputation: Analysis of public opinions regarding Zoom's layoff announcement on social media

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ABSTRACT

Negative organizational events, such as mass layoffs, often trigger intense backlash on social media, thus posing a reputational threat. The viral nature of social media can amplify the concerns associated with a negative organizational event, making it highly visible and attractive to various stakeholders. It is thus imperative for organizations to understand the driving factors behind the negative reactions for effective crisis management. Using Zoom's 2023 layoff as the context of a para-crisis, this study examined publics' responses on social media toward the CEO's authentic communication of the layoff announcement. Latent Dirichlet Allocation (LDA) Topic modeling and VADER sentiment analysis were utilized to analyze 7,611 social media posts. Findings demonstrated that publics responded positively on social media toward the authentic attributes reflected from the CEO's communication. The positive sentiment toward the CEO seemed to have transferred to Zoom's reputation. Practical implications suggest public relations practitioners should involve executives in crisis communication to proactively attenuate reputational damages and analyze online public opinion to obtain a realistic assessment of the reputational threat and the effectiveness of crisis communication.

1. Introduction

Social media has afforded unparalleled power for publics to freely express their opinions and attitudes about organizations (Baker & Hashimoto, 2024; Coombs & Holladay, 2014; Einwiller & Steilen, 2015; Li & Ken, 2021; Zhao et al., 2022). Those opinions and attitudes can serve as a vital information source that can be mined to understand public opinion and generate reputation about a target entity (Benlahbib & Nfaoui, 2020). Given the highly visible and viral nature of social media, public relations scholars (e.g., Huang & Ki, 2023; Sisson & Bowen, 2017) have cautioned the potential risk of negative opinions vented by the publics on social media spiraling into a para-crisis (Coombs & Holliday, 2012, 2014), thus posing a threat to the organizations' reputation (Su et al., 2019). The subject on negative conversations on social media, or negative electronic word-of-mouth (eWOM) (Hennig-Thurau et al., 2004), has increasingly attracted attention from public relations scholars (e.g., Krishna & Kim, 2020; Lee & Kim, 2020; Oh & Ki, 2019; Qu et al., 2024; Xiao et al., 2018; Wang et al., 2022). If negative conversations expressed by the public on social media are not addressed satisfactorily, a para-crisis tends to escalate into a full-blown crisis, possibly incurring substantial reputation damages (Coombs & Holladay, 2012, 2014).

The significant impact of online publics' conversations has prompted communication scholars (e.g., Avery, 2017; Brown-Devlin et al., 2022) to underscore the need to understand these publics' needs and concerns during a crisis. Brown-Devlin et al. (2022) concluded that the emergence of AI (artificial intelligence)-driven big data analytics have transformed public relations practices during crises. Particularly, AI technologies can empower public relations practitioners to analyze public opinion during crises and hence influence positive crisis outcomes. To illuminate the effects of contextual factors that may influence online public opinion in crisis management, communication scholars (e.g., Hearit, 2021; Ndone et al., 2022; Thomsen, 2023; Zhang et al., 2024; Schoofs & Claeys, 2021) have called for examining the role of organizational leaders, such as chief executive officer (CEO), on shaping public opinion. Monehin and Diers-Lawson (2022) lamented that crisis leadership exists as one of the most understudied topics in crisis communication. Given the credibility and symbolic power of CEOs (Clementson, 2020), examining the impact

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of executives' participation in crisis communication can yield crucial insights in crisis management, especially to address the challenge of proactively mitigating potential crisis damages.

In response to the pivotal call from communication scholars to examine how organizational executives' attributes shape public opinion on social media during crisis (Zhang et al., 2024), this study examines how the online public reacts to a para-crisis, specifically, Zoom CEO's communication regarding a drastic downsizing. Specifically, this para-crisis occurred as a result of a large-scale layoff announcement on February 7, 2023 by Zoom, a global video communications company and a hallmark of the COVID-19 pandemic. Zoom's remarkable growth during the pandemic, however, faltered as a result of a saturated market, rising inflation, and intense competition (Ables, 2023). Especially, the technology industry had already been plagued by a landslide of layoffs starting late 2022 (Sucher & Westner, 2023). The rationale of this study to focus on Zoom's 2023 mass layoffs stems from two factors. First, the scale of Zoom's 2023 downsizing, 15 % of its workforce, represented a bigger share of workforce reduction than cuts occurred earlier in prominent tech giants such as Salesforce, Microsoft, and Workday, signaling profound, industry-wide impact (Ford & Bloomberg, 2023). Such a magnitude of impact created a significant buzz on social media, triggering a para- crisis and making Zoom a desirable choice to examine how public reactions transpired on social media. Second, Zoom CEO, Eric Yuan, publicly announced the layoff news on the company's blog-site, 30 min before formal emails reached affected employees, acknowledging mistakes and taking responsibility. Zoom CEO's communication characterized by honesty, transparency, and genuineness represented a distinct departure from other leading tech companies' conventional layoff announcements sent through emails to employees, which appeared *transactional* rather than genuine (McGregor, 2023). These unique attributes surrounding Zoom's 2023 layoff communication thus warrant this current study's focus on Zoom's layoff crisis.

This study centers on a specific leadership communication attribute (i.e., authentic communication), as the Zoom CEO's communication of layoff announcement exemplifies truthfulness, honesty, transparency, and consistency, the core facets of authentic communication (e.g., Shen & Kim, 2012). Although authentic communication has demonstrated positive effects on supportive employee behaviors (e.g., Men, 2021), public engagement of corporate social advocacy initiatives (e.g., Dong et al., 2024), organizational-public relationship outcomes (e.g., Lim & Jiang, 2021), and brand reputation (e.g., Södergren, 2021), there is a limited body of research (e.g., Erickson, 2021; Liu et al., 2017; Monehin & Diers-Lawson, 2022; Waddell, 2020) directly probing the effects of authentic communication during crisis, especially linking organizational executives' authentic communication with online public opinion in crisis management. As online publics' demand for open, transparent crisis communication intensifies (Claeys & Waele, 2022), communication scholars face a pressing need to explore the influence of honest, open, transparent communication on public opinion formed on social media during crisis (Jin et al., 2018; Zhang et al., 2024). An in-depth examination of online public opinion regarding Zoom CEO's authentic communication can thus yield much-needed insights to fill the gaps in the extant scholarship regarding leadership communication and crisis management.

Contextualized in the para-crisis of Zoom's drastic layoff in February 2023, this study seeks to understand the public opinion transpired on social media in response to the layoff communication by the company's CEO. Particularly, this study is built upon two premises. First, the value of authenticity in leadership communication (Jin et al., 2023; Men, 2021; Men & Tsai, 2016) and the role of authentic communication on influencing public opinion during a crisis merit the need to examine the influence of organizational leaders' communication on the public reactions on social media toward a negative organizational event, such as a drastic downsizing. Second, a negative organizational event, such as a sizable workforce reduction from a prominent global organization, is likely to spark intense backlash from the online publics because of the adverse impact on the affected employees and feelings of uncertainty among lay-off survivors. Unsurprisingly, the concerned publics may flock to social media to vent their disappointment or dissatisfaction about the target organization. As reputation is inherently derived from public opinions (Kotras, 2020; Meadows & Medadows, 2016), social discussions occurred as a response to an adverse organizational event (e.g., drastic layoff) can pose substantial threats to the focal organization's reputation (Coombs & Holladay, 2014; Einwiller & Steilen, 2015; Huang & Ki, 2023). It is thus paramount for organizations to understand the reactions from the concerned publics on social media to engage in online reputation management in such tumultuous times.

Given that public opinion is composed of cognitive (e.g., thoughts, views, ideas) and affective (e.g., sentiment) dimensions (e.g., Jungherr et al., 2017; Liu, 2020), this study aims to explore the effects of authentic communication on an organization and its leader's reputation in light of a para-crisis by examining the following: the prominent discussion topics (i.e., cognitive dimension of public opinion) generated on social media about Zoom as a company following Zoom CEO's layoff announcement, (2) the major discussion topics (i.e., cognitive dimension of public opinion) generated on social media regarding the CEO, Eric Yuan, and (3) the sentiments that emerged on social media in response to Zoom CEO's layoff announcement. This study examined a total of 14, 393 social media posts using natural language processing (NLP)-based Latent Dirichlet Allocation (LDA) topic modeling and Valencene Aware Dictionary (VADER) sentiment analysis. Findings of this study are expected to contribute to the rising scholarship on CEOs' authentic communication and online reputation management during a crisis in public relations. This study also yields practical implications for public relations practitioners to involve organizational executives in crisis management and to conduct analyses on public opinion during crises.

2. Literature review

2.1. Downsizing and para-crisis

Organizational downsizing is "an intentional event involving a range of organizational policies and actions undertaken to improve firm performance through reduction in employees" (Datta et al., 2010, p. 282). Strategic choices or decisions such as downsizing may incur damage to an organization's reputation if those choices are not perceived as appropriate in the given context (Schulz & Johann, 2018). Existing studies (Atkins & Favreau, 2022; Hellerstein et al., 2019; Kumar et al., 2023; Stäbler et al., 2023) revealed that downsizing usually incurs negative perceptions from stakeholders due to its adverse impact on the target organization's market value and damage to the stakeholder morale. Specially for employees, layoffs symbolize a breach of moral contract with their organizations (Schulz & Johann, 2018). Unsurprisingly, the announcement of a mass layoff is also bound to trigger damaging perceptions from concerned publics who are likely to turn to social media to voice their opinions regarding the target organization (Coombs & Holladay, 2014; Einwiller & Steilen, 2015; Huang & Ki, 2023). Negative conversations on social media, if not addressed satisfactorily, tend to escalate into para- crises (Coombs & Holladay, 2012, 2014), driven by frustrated stakeholders or publics who seek to make their dissatisfaction loud and clear on social media.

A para-crisis is concerned with a "publicly visible threat" (Coombs & Holladay, 2012) that blames an organization for its irresponsible or unethical behavior (Manika et al., 2017). Para-crises unfolding on social media can pose grave and unpredictable reputational threats to organizations (Zhou & Ki, 2018). Especially, given the viral nature of social media, detrimental posts and conversations on social media can potentially garner sizable social traction, flaming an online firestorm against the target organization (Coombs & Holladay, 2014; Su et al., 2019). It is therefore paramount for organizations to employ appropriate tools to understand the evolving social media reactions from disgruntled publics

to avoid escalating a para-crisis into a full-scale crisis and to attenuate reputational threat from negative organizational events such as mass layoffs (Coombs & Holliday, 2014). When confronted with a reputational threat from a para-crisis, an organization should communicate to a broader range of publics, such as media and the community (Huang & Ki, 2023). With the prevalent use of social media among the public to voice their thoughts and emotions, especially during a crisis, social media platforms thus render themselves a valuable source to examine real-time public opinion. The next section hereby discusses the importance of probing public opinion that unfolded on social media during a crisis.

2.2. Public Opinion on Social Media and Crisis

Public opinion refers to an aggregate of individual thoughts, attitudes, and emotions (i.e., views and sentiments) obtained through retrospective, survey-based methods (e.g., polls) (Jungherr et al., 2017; Lin et al., 2013; Lang & Lang, 2009). The deficiencies of this conventional approach lie in the limited sample size, the challenge to map out real-time public reactions, sensitivity to the phrasing of survey questions, and public distrust of polling (Edwards-Levy, 2021). Ubiquity of social media use has transformed how public opinion is conceptualized and measured (Chen & Tomblin, 2021). Particularly, technological affordances of social media (e.g., mass participation, interactive dialogue, convenient access, immediacy, etc.) have enabled researchers and organizations to capture real-time public opinion through NLP-based data mining techniques (Cinelli et al., 2021). Social media has evolved into "an inherent public opinion infrastructure" (Zhang et al.,2022, p. 4) that shapes the construction, dissemination, and measurement of public opinion. Following a review of existing conceptualizations of public opinion, Zhang et al., 2022) defined public opinion formed on online (e.g., social media) as "naturally occurring, temporally sensitive, and inherently social opinions emerging from the networked publics of social media" (p. 4).

The prevalent adoption of social media has made these digital platforms preferred locations for individuals to share their opinions and seek information during crisis (Lu & Jin, 2020; Morgan & Wilk, 2021; Noor et al., 2024; Yarchi, Baden, & Kligler-Vilenchik, 2021). Real-time, direct feedback from the public during crisis provides organizations with vital insights on relevant stakeholders' urgent needs and threats (Byrd & John, 2022). Negative public opinion during crisis not only poses detrimental impact on organizations' reputation but also threatens their future performance (Chang et al., 2024). The pivotal role of public opinion that formed on social media warrants further research on factors that may influence public reactions during crisis (e.g., Cheng et al., 2017; Hong & Cameron, 2018; Kim et al., 2018; Shah & Schweiggart, 2023; Xie et al., 2022; Xue et al., 2024; Zhang et al., 2024). Especially, communication scholars (e.g., Barkley, 2020; Koch et al., 2022; Liu et al., 2022, Schoofs & Claeys, 2021) have advocated more analyses on the role of organizational leaders (e.g., CEOs) in crisis management. Aligning with such emerging research demand, this study examines online public opinion (e.g., views, sentiment) regarding Zoom CEO's communication about layoff. Specifically, this study investigates the first dimension of online (e.g., social media) public opinion—views and thoughts–through NLP-based topic modeling and analyzes the second dimension of public opinion, namely sentiment, through sentiment analysis. To probe public reactions (e.g., views and sentiment) toward Zoom CEO's layoff communication, the framework of authentic communication is adopted in this study to theoretically contextualize the CEO's communication. Analyzing views and sentiments from social media data has become a widely used approach for studying online public opinion (e.g., Barachi et al., 2021; Chen et al., 2022; Jabalameli et al., 2022; Prichard et al., 2018; Nokkaew et al., 2023; Reyes-Menendez et al., 2018; Zhou, 2022). For example, Dai et al. (2024) employed the LDA topic modeling and sentiment analysis to assess public opinion surrounding the MU5735 flight crash disaster. Similarly, Morgan and Wilk (2021) examined public sentiment on social media regarding the 2018 Cricket Australia ball-tampering scandal by mining the sentiments and themes from social media posts.

2.3. Authentic communication

Defining authenticity. Existing definitions of authenticity have been drawn from ancient Greek philosophy centered on an ideal mix of benevolence, self-knowledge, and ethics (Kernis & Goldman, 2006). Many conceptualizations of authenticity have converged on the theme of obtaining self-knowledge, demonstrating sincerity and genuineness, and acting in congruence with one's values and beliefs (Shamir & Eilam-Shamir, 2018). Authenticity is deemed as one of the most popular leadership buzzwords by both academics and executives (Brodsky, 2022; Jiang & Luo, 2018; Ma et al., 2020; McClimon, 2023; Perkins, 2023; Zhang et al., 2021). There has been a rise in expectation for organizational leaders to strive for authenticity (Jin et al., 2023; Lim & Young, 2021; Men, 2021; Rodriguez et al., 2024; Silva et al., 2023). Although the value of authenticity has been widely embraced, the challenge facing organizations and leaders resides in developing and maintaining authenticity, especially during turbulent times (Fox et al., 2021; Liu et al., 2017).

Based on a comprehensive review of literature related to authenticity in strategic communication, philosophy, sociology, psychology, and business management, public relations scholars Shen and Kim (2012) conceptualized perceived organizational authenticity as a multidimensional construct involving truthfulness/genuineness, transparency, and consistency. Authentic organizations and leaders are true to themselves, transparent, and consistent.

Specifically, truthfulness/genuineness refers to demonstrating a genuine self or acting in accordance with one's true self, which is based on organizations' accurate and honest assessment of themselves and their publics' interests as well as needs. Organizations can cultivate perceptions of genuineness and sincerity through their behaviors and good intentions to publics (Bowen, 2016). The genuineness dimension constitutes "the foundation of transparency and consistency" (Shen & Kim, 2012, p. 375).

The second dimension of authenticity, transparency, is concerned with the need for organizations to transparently demonstrate their genuine selves to the publics (Shen & Kim, 2012). Specifically, transparency entails organizations to willingly assume responsibility for and learn from their mistakes and actions (Kernis & Goldman, 2006). Similar to what Kernis and Goldman labeled as authentic individuals' behavior of unbiased processing, the transparency dimension of authenticity requires organizations to sincerely and objectively acknowledge the consequences of their decisions and behaviors regardless of the positive or negative perceptions from publics (Shen & Kim, 2012).

The consistency dimension of organizational authenticity addresses the issue of whether an organization's actions are aligned with its espoused beliefs, values, and principles (Kernis & Goldman, 2006; Walumbwa et al., 2008). Consistency is contingent on truthfulness and transparency. This is because genuine presentation of itself, impartial as well as transparent evaluation of its actions and decisions enable an organization to behave consistently (Shen & Kim, 2012).

2.4. Authentic communication and CEOs

Given the strategic decision-making role of CEOs and their influence on shaping public opinion and discourse (Barkley & Okamoto, 2023; König et al., 2025; Ostrovsky-Berman, 2021), a rising body of scholarship in public relations has focused on leadership communication (Cho et al., 2023; Ji & Hong, 2023; Jin et al., 2023; Men, 2015). Existing research has examined the role of CEOs' communication content, strategies, channels, and styles in organization-public relationships (e.g., Tsai & Men, 2017), employee engagement (e.g., Men, 2015; Qin et al., 2023), social media engagement (e.g., Men et al., 2018; Thelen & Men,

2018; Yue et al., 2023), crisis communication (e.g., Barkey & Okamoto, 2023; Koch et al., 2022), corporate reputation (e.g., Sauter & Jungblut, 2024), corporate social advocacy and CEO activism (e.g., Cho et al., 2023; Lee & Tao, 2021; Lim & Jiang, 2021), and employee advocacy (e.g., Lee & Dong, 2023).

Most CEOs assume a predominant external role of acting as the "spokesperson" of an organization, explaining an organization's decisions or actions to various stakeholders (Denner et al., 2021; Wu et al., 2022). Such representation by CEOs serves to personify their organization's values, beliefs, and culture to gain desirable perceptions (e.g., image, reputation, brand strength, etc.) from stakeholders (Denner et al., 2019). With the interactive nature and prevalent adoption of social media, CEOs are expected to engage in direct and open interactions with stakeholders on social media platforms (Men & Tasi, 2016; Kim et al., 2016; Yim, 2019). Recent studies in effective leadership communication have consistently identified authenticity as a significant attribute (Dailey, 2021; Men, 2021; Men et al., 2020). Stakeholders are increasingly expecting CEOs to engage in authentic communication to address challenging organizational issues during a crisis (Fan, Zhou, Wang, & Chang, 2020; Laufer, Garrett, & Ning, 2018).

The display of authenticity by organizational leaders when communicating and interacting with stakeholders on social media has emerged as a vital concern (Jin et al., 2023; Men, 2021; Men & Tsai, 2016) with the hyper-personal nature of social media (Walther & Whitty, 2021). As the most visible symbol of an organization, communication from a CEO allows stakeholders to obtain a sense of an organization's values and priorities (e.g., Hambrick & Wowak, 2021; Jin et al., 2023).

Following Shen and Kim (2012)'s conceptualization of organizational authenticity (i.e., genuineness, transparency, consistency), public relations scholars have applied these dimensions of authenticity to examine CEOs' social media communication (e.g., Men & Tsai, 2016), the role of authentic CSR communication on relational outcomes (Lim & Jiang, 2021), the impact of CEOs' communication on employee relational and behavioral outcomes (Men, 2021), CEOs' political tweets (Yim, 2019), and CEO activism (Jin et al., 2023). Drawing from Men (2021) and Lim and Jiang (2021), this study thus defines CEOs' authentic communication as "CEOs' interactions and communications (verbal or non-verbal) with others that reflect truthfulness/genuineness, transparency, and consistency" (Men, 2021, p. 3).

2.5. Authentic communication and corporate reputation

The signaling theory (Spence, 1974) provides a linkage between CEOs' authentic communication and corporate reputation. CEOs (i.e., senders of signals) can disclose relevant information to their publics (i.e., receivers of signals). These signals hence enable publics to differentiate CEOs and the organizations represented by them with distinct qualities (Bergh et al., 2014). The outcomes of a signal are contingent on both the sender's effort and the attributes of the receiver (Zerbini, 2017).

Individual members of publics may make sense of the same signal differently. It is thus worthwhile to study publics' different interpretations of a CEO's communication, such as the online publics' reactions toward a CEO's downsizing announcement.

Corporate reputation, the most salient intangible asset in management, reflects the publics' collective or holistic assessments of an organization's ability to fulfill various publics' expectations and demands (Fombrun, 2005; Harrison et al., 2022). Specifically, such evaluations represent stakeholders or publics' comparisons between the target organization's behaviors and the behaviors of other organizations as well as their various expectations (Deephouse et al., 2016). Assessments of corporate reputation are typically shaped by media reports, special interest groups, opinions, and community leaders (Sisson & Bowen, 2017). Since this study is contextualized in discussions (e.g., conversations, posts, comments etc.) on social media, the concept of online reputation is appropriate. Portmann et al. (2015) defined online reputation as "the publicly kept social evaluation of an entity based on the entities' previous behavior, what was posted by the entity, and what third parties share about the entity on the Internet" (p. 90). Based on the aforementioned discussion, corporate online reputation is inherently shaped by opinions revealed by social media users (Liu, 2020).

Stakeholders have increasingly demanded transparency, openness, and accountability from organizational leaders (e.g., executives) during crises (Jin et al., 2018; Lopes Junior et al., 2024). Particularly, leaders' virtue of "true to themselves," an essential attribute of authenticity, is conducive to instilling a sense of optimism, trust, and confidence among stakeholders during crises (Celik et al., 2016). The link between organizational leaders' authentic communication and corporate reputation during crises has been supported in a limited body of research. Zhang et al. (2024) concluded that organizational executives' participation in crisis communication improved crisis communication effectiveness by fostering positive sentiment, which in return can enhance positive organizational attitudes (e.g., reputation). Rees (2020) stressed that incorporating authenticity in organizations' online communication is pivotal to shape desirable perceptions on corporate reputation, especially during a crisis. Shirdastian et al. (2019)'s examination of Twitter conversations on Starbucks evidenced a clear association of this coffee brand's authentic communication with social media users' positive sentiment. By examining the communication from two CEOs in press conferences during crises, Ugoalah (2015) revealed authentic communication from CEOs was effective in shaping positive public perceptions and responses toward the crises, which ultimately had a favorable impact on the organizations' reputation. Kim and Krishna (2018)'s study discovered that South Korean citizens' perceptions of authenticity from government communication significantly affected positive public sentiment (e.g., public engagement) and reduced negative public sentiment (e.g., cynicism).

Given the paucity of extant research on CEOs' authentic communication and corporate reputation during crisis, this study explores the connection between a CEO's authentic communication and the company's reputation, as reflected in public opinion on social media in response to Zoom CEO's communication about mass layoffs. This study thus proposes the following main research question:

RQ1: How was the reputation of the company and its CEO discussed on social media in response to the authentic communication of the layoff announcement by the CEO of Zoom?

Specifically, the assessment of reputation influence from the Zoom CEO's authentic communication is based on examining the discussion topics and sentiment expressed by social media users through the LDA topic modeling and VADER sentiment analysis. As noted earlier, corporate reputation is inherently derived from public opinion. Examining public opinion on social media about the target company, Zoom, and the CEO in the wake of its CEO's communication of layoff, appears to be a logical choice to probe the effects of a CEO's authentic communication on reputation. The examination of social media discussions surrounding Zoom CEO's authentic communication offer practical insights into whether reputational damage to an organization and its leaders might be mitigated in the midst of a layoff crisis. As noted earlier, public opinion is composed of cognitive (e.g., views or thoughts) and affective (e.g., sentiment) dimensions (Liu, 2020). Therefore, the following sub questions are proposed to probe the reputational effects resulting from the cognitive aspects of public opinion generated on social media as a response to Zoom CEO's authentic communication of layoff:

RQ1a: What topics were discussed on social media about Zoom following the authentic communication of the layoff announcement from the CEO of Zoom?

RQ1b: What topics were discussed on social media about Zoom's CEO following the authentic communication of the layoff announcement from the CEO of Zoom?

2.6. Organizational leaders' authentic communication and public sentiment

As part of public opinion, online public sentiment indicates collective attitudes or feelings manifested in narratives expressed on digital platforms in reaction to a particular issue or crisis, which typically imply a positive or negative sentiment (Liu, 2020). The abundance of user-generated content on social media provides valuable insights for analyzing the public's emotional responses during a crisis (e.g., Myint et al., 2024; Scotland et al., 2024). Tracking and understanding the evolving the public's emotional reactions during crises is crucial for organizations to identify the drivers of sentiment shifts, thus gaining deeper insights into the diverse needs and concerns of key stakeholders (Morgan et al., 2021; Shah & Schweiggart, 2023; Zhang et al., 2024). Especially, the contagious nature of emotions serves as a dominant impetus driving activities (e.g., circulation of public opinion) of social media users during crises (Ma et al., 2024; Wang & Zhang, 2024).

This study joins a rising body of scholarship that examines public sentiment surrounding authentic communication displayed on social media. Authentic communication characterized as being transparent, genuine, and consistent is effective at fostering positive sentiment or emotional reactions. For example, Authentic communication by tourism social media influencers significantly affected consumers' positive perceptions toward the endorsed destinations (Dong et al., 2023). Södergren (2021)'s systematic review on brand authenticity concluded that perceived brand authenticity generates desirable brand reputation evaluations and emotions. Studies (e.g., Papadopoulou et al., 2023; Guèvremont & Grohmann, 2018) found that communication centering on brand authenticity can provide a significant buffering factor to foster brand forgiveness and attenuate the negative consequences during crises (e.g., brand scandals). Nayani et al. (2022)'s study revealed that organizational communication focusing on authenticity-building (e.g., demonstrating care and genuineness) during the COVID-19 crisis fostered positive emotions (e.g., trust, supportiveness) among employees regarding workplace accommodations and changes.

Although organizational leaders' authentic communication has received ample scholarly attention (e.g., Anders, 2021; Dailey, 2021; Liu et al., 2017; Men, 2021; Men & Tsai, 2016; Yan et al., 2017; Westfall, 2018), the body of research that examines the direct linkage between authentic communication on public sentiment during crisis is still at its nascent stage. Based on the current research on the positive influence of authentic communication during a crisis, it is reasonable to deduce that authentic communication from organization leaders (e.g., CEOs) tends to elicit positive public sentiment on social media during crisis. Aligned with this logic and to fill the void in studying executives' authentic communication and public sentiment, this study seeks to explore the reputational effects from public sentiment (i.e., the affective dimension of public opinion) regarding both the company and its CEO through the following research question:

RQ1c: What sentiments were expressed on social media in response to the CEO of Zoom's authentic communication about the layoff announcement?

To answer the above proposed research questions, this study utilizes two NLP-based techniques: topic modeling and sentiment analysis. A brief review of these two computational data-mining techniques is thus provided below.

2.7. Sentiment analysis

Sentiment analysis, a natural language processing (NLP) technique, refers to the automatic process of subjective and emotion-laden opinions expressed in natural languages collected from social media platforms (Puschmann & Powell, 2018). A standard procedure in sentiment analysis focuses on extracting polarity categories of emotions from texts, determining whether a text reveals positive, neutral, or negative sentiment towards the target entity (Bonta et al., 2019). The VADER (Valence Aware Dictionary and Sentiment Reasoner) model is widely adopted for the purpose of identifying polar sentiments (Hutto & Gilbert, 2014). The VADER sentiment tool relies on a lexicon-based (i.e., dictionary-based) and rule-based approach to classify texts as positive, negative, or neutral sentiment (Veena et al., 2021). Based on a text's proximity to the positive, neutral, or negative sentiments, a weighted value is computed to demonstrate the intensity or strength of the sentiment conveyed in the text. The reliability and accuracy of VADER in categorizing sentiments from natural languages have been consistently supported in social media analytics research (e.g., Agarwal, 2019; Elbagir & Yang, 2019; Hoti & Ajdari, 2023; Singh et al., 2022).

There is a growing body of research utilizing sentiment analysis to examine perceptions of reputation from online publics. Mishra and Sharma (2019) stressed the importance of detecting emotions through VADER sentiment analysis from social conversations on Facebook regarding Nestle's Maggi noodle crisis, which could help develop communication programs centered on trust to restore reputation. Through sentiment analysis, Calissano et al. (2020) identified key reputation drivers (e.g., food quality, technology innovation, cost, architecture of the pavilions, etc.) for one of the largest events in Europe, Expo Milano. Yuniarti and Ruldeviyani (2023) conducted sentiment analysis of bank customers' comments on social media to identify the reputation risk factors (e.g., mobile application errors, ATM service complaints, etc.). Zammarchi et al. (2023) applied VADER sentiment tool to assess twitter users' opinion on Italy's country reputation as a result of the COVID-19 outbreak.

2.8. Topic modeling

Both topic modeling and sentiment analysis are two key applications in NLP to study online public opinion. Extracting and analyzing unstructured social media content through NLP-based topic modeling has emerged as a new, powerful means to study near real-time public opinion during crises (Fan et al., 2021; Lock & Pettit, 2020). Based on iterative algorithms generating semantic associations among texts, topic modeling aims to extract meanings from large corpus of textual data (e.g., social media conversations) through identifying recurrent topics (Dahal et al., 2019). Classified topics are typically viewed as patterns across datasets. Topic modeling has been widely used to summarize what the public is concerned about from social media data during crisis (Deng et al., 2020). Insights generated from topic modeling can serve as actionable pieces of information during crisis management (Page et al., 2023; Gao et al., 2024; Wang et al., 2024; Zhou et al., 2024).

This study specifically utilized Latent Dirichlet Allocation (LDA), the most widely applied technique for topic derivation (Nugroho et al., 2020; Yu et al., 2023). First introduced by Blei et al. (2003), LDA is a probabilistic generative model to extract the most salient topics relying on co-occurrence of words from textual data through unsupervised machine learning (Westrupp et al., 2022). Specifically, LDA utilizes algorithms to discover how the keywords cluster statistically, indicating the compatibility between words and topics in the texts. LDA represents a reliable public opinion technique as it discovers nuances and relationships in the data (Nelson, 2020) as well as non-obvious connections between ideas expressed in large volume of textual data that may not be identified through manual coding (Saura et al., 2023; Weiss & Muegge, 2019).

To determine the optimal number of topics, we used Grid Search in combination with the Coherence Score. Grid Search is a technique used for hyperparameter optimization technique that systematically tests different combinations of predefined hyperparameters to identify the best-performing configuration on the validation set. For the research, we conducted a Grid Search by experimenting with various values for the number of topics (e.g., 5, 10, 15, 20, etc). The Coherence Score is another indicator used to evaluate the quality of topic models, such as Latent Dirichlet Allocation (LDA). It measures how consistent and semantically related the words within a topic are. A higher Coherence

Score indicates better quality of the topic model, as it reflects more consistent and cohesive keywords within each topic.

3. Method

Utilizing NLP techniques, this study analyzed 7611user generated social media posts through topic modeling and sentiment analysis. Latent Dirichlet Allocation (LDA) topic modeling identifies thematic trends in the large corpus of textual data to reveal major topics from the social media discussions regarding a specific topic. Valence Aware Dictionary and Sentiment Reasoning (VADER) detects positive, negative, and neutral sentiments. Using these NLP-based data mining techniques can provide deeper insights into understanding the public's responses to Eric Yuan's communication of Zoom's mass layoffs and its consequences for Zoom as a corporation.

Following Grunig (1992)'s conceptualization, this study views publics as any individuals, groups, organizations, or entities that are affected or can affect an organization. Hence we treated both individual users and organizational accounts on social media as part of the broader social media publics responding to the Zoom CEO's layoff communication. Reactions (e.g., likes, share, comments) from organizations on social media can represent institutional perspectives and were thus deemed as relevant. Aligned with this orientation, this study considered responses from both individuals and organizations on social media as valid expressions from the broader stakeholder publics as they represented a range of online perspectives and voices.

3.1. Data collection

Zoom's CEO, Eric Yuan, shared an announcement regarding layoffs on Zoom Blog on February 7, 2023 and also posted the link to this announcement on X (formerly Twitter). In order to capture the public's responses to Zoom's layoffs, several search queries were built. In order to understand the public's opinion on Zoom, the following search was built: "("Zoom") AND ("layoffs" OR "layoff" OR "downsizing" OR "employees" OR "job cuts" OR "lay off" OR "tech layoffs" OR "#layoff") NOT ("zoom call" OR "zoom conference")." To gather social media posts related to Eric Yuan, Zoom's CEO, the following query was created: "("Zoom CEO" OR "CEO of Zoom" OR "Eric Yuan" OR "@ericsyuan") AND ("layoffs" OR "layoff" OR "downsizing" OR "employees" OR "job cuts" OR "lay off" OR "tech layoffs" OR "#layoff") NOT ("zoom call" OR "zoom conference")." The terms "zoom call" and "zoom conference" were removed from both search queries to avoid collecting posts related to meetings.

A total of 14,383 social media posts in the U.S. region were initially retrieved using a social media listening platform and tool, *Brandwatch*, for a span of a month beginning on the day the Zoom CEO made his layoff announcement (i.e., February 7, 2023 to March 7, 2023). Social media posts were gathered from Facebook, YouTube, X, and Reddit. Specifically, 11,118 social media posts were collected for the Zoom corporation with 7665 posts (i.e., 68.98 %) from X (i.e., tweets, retweets, and replies), 3323 posts (i.e., 29.88 %) from Reddit, 107 posts (i.e., 0.96 %) from Facebook, and 23 posts (i.e., 0.21 %) from YouTube. Of those posts, 6509 posts were original posts, 4344 posts were retweets, and there were 265 replies. A total of 3265 social media posts for the CEO of Zoom were initially retrieved for the same period of time with 2540 posts (i.e., 77.80 %) from X, 701 posts (i.e., 21.47 %) from Reddit, 22 posts (i.e., 0.67 %) from Facebook, and 2 posts (i.e., 0.06 %) from YouTube. Of those posts, 1476 were original posts, 1654 were retweets, and 135 posts were replies. The high distribution of social media posts on X echoed existing research (e.g., ; Kinsky et al., 2021; Noor et at., 2024) that X emerges as a dominant social media platform in crisis communication. All social media posts were mined in the English language and cleaned prior to analysis.

3.2. Data cleaning

Preparing the data is a critical step to ensure the integrity of data analysis. By removing unwanted information and elements from the raw data that could potentially skew or obscure the analysis, researchers can yield more accurate and meaningful outcomes. Therefore, any posts that were irrelevant or disruptive for textual analysis should be removed prior to analyzing the corpus (Hickman et al., 2022). Along with the tokenization process where sentences and phrases are split into smaller units or tokens, special characters, emojis, hyperlinks, stop words ("insignificant" words that provide low context such as "the," "of," "a," etc.) were removed.

Specifically, both the Zoom and Zoom CEO datasets contained a large number of posts mentioning "employees," many of which were unrelated to the Zoom layoffs. To address this, this study implemented a rigorous process, applying additional measures to ensure the removal of all irrelevant data before conducting the final analysis. Only posts explicitly referencing "employees" in the context of Zoom layoffs were retained. Furthermore, meticulous steps were taken to ensure that data only related to the Zoom brand was not misclassified into the dataset for the Zoom CEO and vice versa. After a thorough cleaning process to remove misclassified texts, 351 posts were removed from the Zoom data, and 143 posts were deleted from the Zoom CEO data.

There were naturally overlapping posts mentioning both Zoom as a company and the announcement of the Zoom CEO that should be acknowledged. Exactly overlapping posts, where full text from the Zoom data are identical with the full text of the Zoom CEO data, were identified to be 528 posts. Additionally, we normalized the full text and took similarity scores of texts to see how similar the texts across both datasets were. If the similarity score was greater than 80, we deemed those texts to also overlap across the datasets. Similarly overlapping posts were identified as 308 posts. However, it should be noted that these overlapping posts were kept in the dataset because they were not misclassified data. Rather, they were posts that discussed both Zoom as a company and the deeds of its CEO.

Aside from misclassified posts and overlapping posts, additional posts that did not have relevance to Zoom layoffs, but only Zoom (e.g., Zoom meeting links, Zoom products, etc.) were removed. After a thorough cleaning process, a total of 4837 posts were removed from the Zoom dataset, and 1441 posts were removed from the Zoom CEO dataset. Therefore, the finalized dataset for Zoom resulted in 5930 posts and 1681 posts for the Zoom CEO.

3.3. Data analysis: topic modeling and sentiment analysis

Topic modeling through LDA analysis allows the identification of underlying thematic structures within a textual corpus (Diesner & Carley, 2010; Park & Choi, 2023). Using the Gensim library, which is specifically designed for unsupervised topic modeling, the parameters were carefully chosen for this study's analysis. The LDA model is probabilistic, exploring different combinations of words and documents to identify the most coherent topics, in this case, strings of five terms. The extracted topics display the highest probability of a group of words (five) that occur together to present the most frequently discussed topics among the large textual corpus.

It is necessary to explain how topic words and associated descriptions are determined by the LDA model. Topic models output a set of high-probability words that frequently occur together which help define the topic. As a probabilistic model, the LDA model was set in this study to automatically generate a string of five words that occurred most frequently together, along with readings of available tweets under each topic to understand the topic's context. When interpreting topics derived from LDA, descriptions were created by understanding the context of the top words identified for each topic. Specifically, in addition to the specified words that were identified, we utilized thematic analysis (Braun & Clarke, 2022) to generate themes emerging from the

social media posts that contained the identified key words in each topic. These themes thus help us to understand the context and the nuanced meaning that the identified key words in each topic convey. The thematic analysis serves as a thorough and interpretive process for uncovering emerging themes and underlining the association among key semantic elements in texts (Tranfield et al., 2003). As it is not realistic to examine every social media post included in each major topic, both authors examine posts associated with each identified topic to identify a theme. Specifically, a theme "captures something important about the data in relation to the research and represents some level of patterned response or meaning within the data set" (Braun & Clarke, 2006, p. 82). Following this definition, a common theme shared in the posts represents an interpretation of meaning from the examined posts.

Particularly, the authors followed the six-step process for thematic analysis developed by Braun and Clarke (2006): (1) familiarization with the data through repeated reading and documenting initial ideas; (2) generating initial codes by organizing data into meaningful groups; (3) searching for potential themes by clustering codes; (4) reviewing and refining candidate themes; (5) defining and naming themes; and (6) producing the thematic description. We manually coded the representative posts and used Excel sheet to organize codes and track emerging patterns across posts. Each author generated themes from the posts associated with each major topic and compared coding results, discussed differences, and reached consensus on the final themes throughout the coding and interpretation process to ensure validity and trustworthiness. Furthermore, the traceable documents (e.g., original data, notes, etc.) and frequent discussions between the authors also enhanced the credibility and confirmability of our thematic analysis (Nowell et al., 2017).

VADER was chosen to assess the sentiment of this study as it is known to be more sensitive particularly to sentiment expressions in social media contexts (Elbagir & Yang, 2019). The VADER sentiment analyzer classifies data as positive, negative, neutral or compound. The compound value is a metric used to measure sentiment in a given data. For this study, a compound value greater than 0.05 is considered positive, while a compound value between −0.05 and 0.05 is considered neutral, and a compound value below −0.05 is considered to have negative sentiment. Additionally, social media posts are classified into a frequency distribution of positive, negative and neutral words to provide further insight into sentiment.

3.4. Content analysis for Zoom CEO's announcement

The central focus of this study is to examine how online social media users reacted to Zoom CEO Eric Yuan's authentic communication of the layoff announcement. To ensure that the CEO's layoff communication met the defining characteristics of authentic communication, a content analysis was conducted on the announcement. The two authors who were well trained in the literature on authentic communication coded the CEO's layoff announcement obtained directly from the company's blogsite. Each sentence in the Zoom CEO's layoff announcement was coded for its reference to the three core dimensions of authentic communication as conceptualized by Shen and Kim (2012) as well as Lim and Jiang (2021): truthfulness/genuineness, transparency, and consistency. Subcategories were created for each dimension of authentic communication based on the definition (Shen & Kim, 2012). First, the truthfulness and genuineness dimension of authentic communication is evaluated through four subcategories: (1) whether the company's layoff was communicated in a genuine manner; (2) whether honest and complete information was provided about the company's layoff; (3) whether the CEO conveyed the layoff sincerely; and (4) whether the CEO told truthful stories about the reasoning behind the layoff decision.

Secondly, the transparency dimension of authentic communication is assessed by (1) whether the CEO acknowledges mistakes and (2) whether the CEO accepts and learns from his mistakes. Lastly, the third dimension of authentic communication, consistency, was evaluated by (1) whether the CEO matches his rhetoric with his action; (2) whether the CEO demonstrates consistency between his beliefs and actions; and (3) whether the CEO's behavior matched the company's core values.

For these above-mentioned subcategories for each dimension of authentic communication, the presence was coded as 1 and absence as 0. Then the total was added across the subcategories of a specific dimension to represent the score along that dimension. Inter-coder reliability was assessed using Krippendorff's α with values ranging from.84 to 1.00, indicating strong agreement among coders (Hayes & Krippendorff, 2007).

4. Results

RQ1: How was the reputation of the company and its CEO discussed on social media in response to the authentic communication of the layoff announcement by the CEO of Zoom?

To study public opinion on social media regarding Zoom CEO's authentic communication of the layoff announcement, two NLP-based techniques, namely topic modeling and sentiment analysis, have been employed to provide structure and meaning to the large textual corpus that is meaningless in its original, unstructured form otherwise. The main research question explores how the authentic communication of the layoff announcement from Zoom's CEO affected the reputation of the company and its CEO. This study will address this main research question through subsidiary research questions. Particularly, as reputation is intrinsically shaped by public opinion, identified topics and sentiment serve as indicators on how publics on social media viewed the reputation of Zoom as well as its CEO in light of the CEO's authentic communication during this layoff para-crisis. Specifically, the first two sub research questions used LDA-based topic modeling to examine the cognitive aspect of online public opinion toward the company and its CEO through the major discussion topics expressed by social media users in response to Zoom CEO's authentic communication of the layoff announcement. The major discussion topics and sentiment expressed by social media users about the company and its CEO as reactions to the CEO's layoff communication can provide insights on the perceived reputation of the company and its CEO. To answer the third sub research question, VADER sentiment analysis was employed to examine the sentiments expressed by social media users regarding Zoom and its CEO in light of the CEO's authentic communication of the layoff announcement.

This study aims to examine the social media users' reactions toward Zoom CEO's layoff communication, with the underlying premise that the communication was authentic. To establish this authenticity, we first conducted a content analysis of the CEO's layoff announcement. The results confirmed that the layoff communication aligned with the defining characteristics of authentic communication as conceptualized in the literature (e.g., Shen & Kim, 2012). For example, the CEO's statement –" We have made the tough but necessary decision to reduce our team by approximately 15 % and say goodbye to around 1300 hardworking, talented colleagues"—reflected truthfulness in the CEO's communication, openly acknowledging the difficult decision and its personal and organizational impact. The CEO's admission of "reducing my salary for the coming fiscal year by 98 % and foregoing my FY23 corporate bonus" clearly demonstrated his willingness to be held accountable (i.e., transparency dimension of authentic communication) for the company's decision to downsize drastically and showed consistency between his words (e.g., apology) and actions (e.g., punitive actions for mistakes). Additionally, the CEO's communication echoed the company's core value of *care* (as outlined on Zoom's official website) in instances such as "I want to focus on what matters most; you," or "Impacted employees will also receive outreach from a leader in your organization offering to have a 1:1 check-in should you want it; they are here to offer support during this transition." Such correspondence with the company's core value manifested the consistency dimension of authentic communication as specified in the literature (Shen & Kim, 2012). Results from this content analysis established the necessary

context to probe online public reactions to Zoom CEO's authentic communication of the layoff announcement, which are presented below.

RQ1a: What topics were discussed on social media about Zoom following the authentic communication of the layoff announcement from the CEO of Zoom?

Three most prevalent topics were identified through topic modeling of 5930 social media posts collected for the Zoom corporation. Topic models output a set of high-probability words that frequently occur together which help define the topic. Three dominant topics regarding the Zoom company were determined by the LDA model.

Interestingly, results from topic modeling revealed that conversations on social media were dominated by recent mass layoffs in the technology industry at large. Discussions were not specifically focused on Zoom layoffs but rather, zoom layoff was mentioned as an example of one instance of layoffs in the wave of a series of layoffs in the technology corporations from Microsoft, Dell, Amazon, Salesforce, eBay, IBM, Gitlab, GoDaddy, Disney, Google, etc. This topic was derived from the most frequently occurring groups of words within the dataset, attributing to almost half of the collected dataset (3092) on the Zoom brand, represented by a string of the following five words, 'jobs,' 'microsoft,' 'amazon,' lay,' 'tech.' Social media users shared and retweeted media reports and commentaries from legacy media outlets such as *Forbes*, *NPR*, *New York Times*, *The Guardian*, *The Washington Post*, and others that reported on the numbers of serial layoffs in the technology workforce such as posts highlighting statistics and numbers of recent jobs lost (i.e. "Disney 7000 Zoom 1300 GoDaddy 530 EBay 500 Affirm 480 (19 % of workforce) Chase (mortgage) 500 Morgan Stanley (mortgage) 300").

The second most discussed topic was concerned specifically with Zoom's announcement of firing 1300 employees, which is equivalent to 15 % of the total workforce. Conversations pertaining to this topic focused on linking the layoff announcement with the exit of Zoom's president (Greg Tomb), reminiscing upon Zoom's previous pandemic glory (i.e. "pandemic-fueled surge in demand for digital services wanes"), expressing doubts on Zoom's future, reflecting on missed signs of trouble (e.g., the company's reduced meeting schedules), offering layoff advice (e.g., networking, seeking new opportunities, self-care, etc.), and sharing personal layoff stories. Many social conversations used direct quotes from Zoom CEO's announcement about the layoff reasons to discuss the rationales for Zoom's decisions for the mass layoffs, suggesting that the social media publics agreed with the reasons and resonated with the emphatic tone offered by the CEO. Eric Yuan's salary pay cut was also discussed in line with general CEOs taking pay cuts (i.e. "cuts in executive pay," "other executives (will also) take a significant pay cut").

Thirdly, Zoom's layoff made headlines and was discussed at the forefront as an example of the recent layoff "trends" in the technology industry. Conversations about these industry-wide layoffs speculated on the gloomy economic prospects and potential of future layoffs, compared the scale (i.e. percentages) of each layoffs, discussed possible repercussions (e.g., recruitment freeze, stock prices, stability of big corporations, etc.), and shared coping strategies for job loss. A different set of companies are discussed than the mega corporations mentioned in topic #1, such as Bark, Twilio, March Rigetti Computing, IRobot, Rivian Auto, etc.

RQ1b: What topics were discussed on social media about Zoom's CEO following the authentic communication of the layoff announcement from the CEO of Zoom?

The data collected for Eric Yuan, Zoom's CEO, was smaller compared to the data gathered about the Zoom corporation itself. Among the 1681 social media posts, based on LDA topic modeling three prominent topics emerged. The most notable topic (n = 747 posts) among online public discussions centered on Eric Yuan's admission in his layoff announcement to significantly reduce his salary and forgo his bonuses (topic 1). Social media users acknowledged Eric Yuan's voluntary decision to cut his salary and bonuses through words such as "take responsibility," "cut his own pay," "not taking bonus," and "that's leadership."

Table 1
Major Topics on Social Media for Zoom Company.

Topic Id	Topic	No. of tweets	Description
1	'jobs,' 'microsoft,' 'amazon,' lay,' 'tech'	3092	Number of employees affected by mass layoffs in the tech sector, such as Google, Amazon, Dell, Zoom, Microsoft, Salesforce, Disney, Gitlab among others are discussed.
2	'pay,' 'lay,' 'tech,' 'latest,' 'staff'	1630	The latest details of Zoom's layoff announcement are discussed along with layoffs of 1300 employees equating to 15 % of its staff and cuts to Eric Yuan's pay.
3	'zm,' 'reduction,' 'workforce,' 'us,' 'mass'	1208	News of Zoom's mass layoff as "Zoom reductions" are highlighted at the forefront of recent rounds of layoffs that are trending in the tech workforce/industry

Table 2
Major Topics on Social Media for Zoom CEO.

Topic Id	Topic	No. of tweets	Description
1	'salary', 'pay', 'laying'", 'cuts', 'zm'	747	Zoom CEO's layoff announcement is discussed with a focus on Eric Yuan's (i.e., the Zoom CEO) salary pay cut and foregoing of bonuses.
2	'salary', 'year', 'said', 'base', 'new'	549	CEO's new base salary in the upcoming fiscal represents a 98 % cut dropping from $301,731 to $6034.62 along with a 20 % cut in the executive team's new base salary.
3	'news', 'salary', 'stock', 'show', 'today'	267	CEO's salary reduction is discussed along with the news about an increase in Zoom's stock prices, attributing to value of Zoom CEO's shares to increase by $246,000,000.

Another prominent topic that gained traction in social media discussion about Zoom's layoffs focused on the substantial salary reduction of the company's CEO who voluntarily slashed his base salary by 98 % from $301,731 to $6034.62. This significant reduction, along with a 20 % cut in the new base salaries of Zoom's executive leadership team, was interpreted by online discussants as an effort by Zoom leadership to take accountability for the company's missteps. Many online users acknowledged and even commended the CEO's self-imposed financial penalty, especially given that executives rarely waive their own compensation during mass layoffs. Despite reducing the CEO's salary substantially, not all online publics were impressed. Some critics dismissed the CEO's pay reduction as "a symbolic cut as he's a billionaire and most of the gains in stock price occurred after this was announced yesterday."

The third most discussed topic was CEO Eric Yuan's salary cut and its impact on the surge of Zoom's stock prices. The Zoom CEO's admission of mistakes and voluntary pay reduction were prominently mentioned in this cluster of social discussions. The leader's move to take responsibility for this mass layoff was recognized in the social posts and mentions through the use of words such as "making sacrifices," "taking personal responsibility," "taking a cut," sharing responsibility," "apology," "admitting mistakes," and "slashing own salary." Notably, investors seem to gain confidence despite Zoom's layoffs due to the CEO's authentic communication. Beside genuineness and transparency, Yuan translated his apologetic words into action by slashing his salary by $295,000, which manifested the transparency (e.g., taking responsibility) and consistency (e.g., aligning words with actions) dimensions of authentic communication. On the cynical side, others also

speculated the significance of his salary reduction. According to Forbes (Kelly, 2023), share prices of companies that announce layoffs, such as Microsoft, Meta, and Amazon, usually increase as employee compensations for tech employees often come from stocks. On the surface, it appeared as if Zoom's CEO held himself accountable by forfeiting his bonus and taking a pay cut, but a significant part of executives' compensations is generated from stock rewards (Kelly, 2023). The public discourse echoed this as it was apparent that cutting "his own salary by $295,000 shows that he's making sacrifices too." However, the CEO's salary cut was compensated for by the rise in Zoom's share price as "the value of his Zoom shares increased by $246,000,000 today as the stock surged on news of the layoffs."

RQ1c: What sentiments were expressed on social media in response to the CEO of Zoom's authentic communication about the layoff announcement?

Sentiments expressed on social media conversations can serve as indicators of corporate reputation, providing insights for online reputation management (Benlahbib & Nfaoui, 2019; Tavazoee et al., 2020). Mining the sentiments expressed on social media from publics regarding Zoom's drastic layoff in 2023 thus can generate insights on Zoom's online reputation indicated from social media discussions. According to the VADER sentiment analysis (Hutto & Gilbert, 2014), a compound value greater than 0.05 is considered positive, while compound values below −0.05 is considered to be negative. The overall sentiment for Zoom was moderately positive with a compound score of 0.07 in our VADER sentiment analysis despite the disastrous nature of mass layoffs. The overall sentiment for Zoom was neutral with a positive lean as the average positive score was 0.043. The overall sentiment reflected in the text is neutral to slightly positive, indicating a generally balanced tone with a mild inclination toward positivity. CEO average compound score is 0.3604, it indicates an overall positive sentiment in the analyzed text. While not highly positive, it suggests a generally optimistic or favorable tone.

To obtain a more nuanced understanding of the sentiments for the target organization,

Zoom, and its CEO–Eric Yuan, we performed separate VADER analyses on the weekly sentiments expressed on social media. Based on the average weekly sentiment scores from VADER, we drew a sentiment chart to demonstrate the change of sentiment on a weekly basis for Zoom (see Fig. 1) and Zoom CEO (see Fig. 2) respectively. Neutral sentiment was purposely removed in generating the sentiment figures displayed below to highlight the evolvement of positive and negative sentiments over the observation period of February 7 to March 7, 2023.

Noticeably, positive sentiment toward the Zoom company consistently outweighed negative sentiment during the observation period of February 7 to March 7, 2023. Particularly positive sentiments toward Zoom underwent a significant surge following the CEO's layoff communication on February 7, climbing to the highest point of positive sentiment in the third week.

Interestingly, the most discussed topic (i.e., Topic #1: 'jobs,' 'microsoft,' 'amazon,' 'lay,' 'tech') for Zoom occurred primarily during the first week (i.e., February 7 - February 12, 2023). A closer examination of social media posts expressing strong negative sentiment during this period reveals that public criticism was mostly centered on the large number of employees affected by the layoffs. When mentioning these series of layoffs, online publics used negative words such as "irresponsible," "brutal," "cruel," "painful," "greedy," "sad" among others to express their dissatisfaction. However, Zoom, as an organization, was not the primary target of criticism among the negative sentiments. Despite the layoffs, positive sentiment was stronger than negative sentiment in the first week and continued to rise through the second and third week. In the final week of observation (i.e., March 1 - March 7, 2023), both positive and negative sentiments declined. Merely 22 social media posts about Zoom were retrieved in our data set during this last week, reflecting a reduction in online discussions and decreasing public interest over time. Notably, negative sentiment in this final week focused on Zoom's abrupt termination of its president, Greg Tomb, on March 3, 2023. Online publics voiced confusion over this sudden firing, as Zoom provided little explanation. However, given the low volume of data in the last week, these sentiments may not fully represent broader public opinion during that time.

Similar to the distribution of sentiments for the Zoom brand, positive sentiment toward the Zoom CEO clearly topped over negative sentiment throughout the one-month observation period (i.e., February 7 - March 7, 2023). Although the average sentiment scores fluctuated, the overall trend exhibited a clear dominance of positive sentiment over negative sentiment. During the first two weeks, positive sentiment largely stemmed from public recognition and appreciation for CEO Eric Yuan's leadership accountability. Social media discussions underscored his willingness to deliver a sincere and genuine layoff announcement, admit mistakes, and take pay cuts.

Markedly, the average positive sentiment scores for Zoom CEO were consistently higher than those scores for the Zoom corporation. Considering the frequent acknowledgement of Zoom CEO's taking

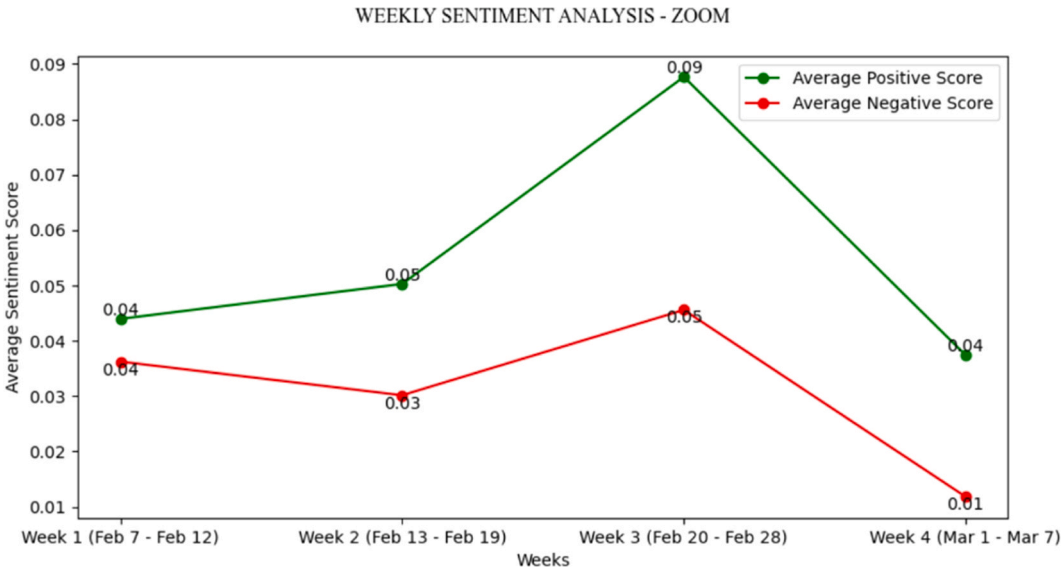


Fig. 1. Weekly Sentiment Analysis for Zoom.

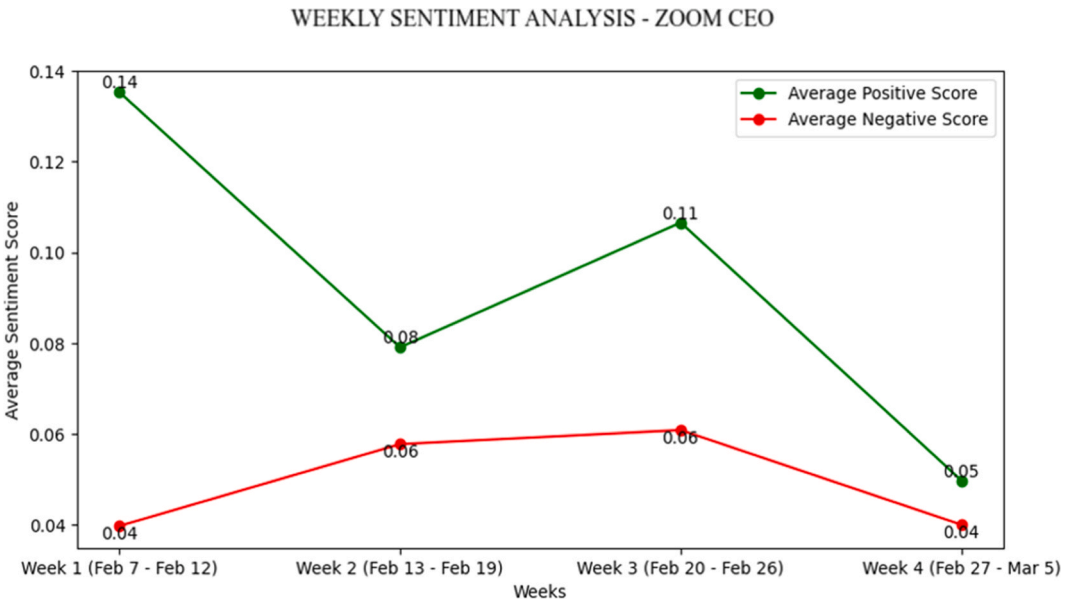


Fig. 2. Weekly Sentiment Analysis for Zoom CEO.

accountability, making self-sacrifice, show care and empathy, as well as admitting mistakes in the positive social media posts,such difference indicates a favorable emotional leaning toward the CEO. The gap between positive and negative sentiment scores was also more pronounced for the Zoom CEO compared to Zoom, especially during the first three weeks. This disparity, coupled with the recurring recognition of the CEO's leadership accountability, empathetic approach, and admission of mistakes in the layoff announcement shown in the positive posts suggests a generally favorable public sentiment in response to Zoom's layoffs.

Interestingly, social media posts voicing positive sentiment also reflected sympathy, concern, and support for the affected employees. Many users shared their own layoff experiences and provided advice on networking, self-care, and career resilience. Impressively, some former Zoom employees affected by the downsizing expressed gratitude for their time at the company, felt proud of being a "Zoomie" (as Yuan referred to employees as "Zoomies"), and still maintained a positive view of the organization despite being laid off. This positive sentiment also seemed to have extended to Zoom as a corporation, influencing its overall perception. Naturally, the sentiment for a corporation that just cut down 15 % of its employees was expected to have strong negative sentiment. However, Yuan's layoff communication seemed to have neutralized the backlash, as the overall sentiment for Zoom was neutral to slightly positive based on the collected data.

Examining social posts during this one-month period revealed the following themes underlying the negative sentiment expressed by online publics. First, online publics expressed doubts whether the CEO's salary cut was truly significant given his lucrative stock bonuses. Second, CEOs from other big technology companies were shamed for not being held financially responsible for layoffs, fueling broader frustration. Third, online publics expressed fear and uncertainty about their own job stability amidst the widespread layoffs in the tech industry. Fourth, the substantial workforce reduction at Zoom was interpreted as a leadership failure. Fifth, social discussions reflected uncertainty and doubts about Zoom's future success. Lastly, social conversations revealed frustration and disappointment over the collapse of the digital technology boom amidst the sharp decline of the stock prices for technology companies.

5. Discussion

The empowering nature of social media has afforded publics an open platform to voice their concerns and needs. Negative discussions by publics on social media are likely to create a para-crisis for organizations, posing a highly contagious and visible reputational threat. Public relations scholarship has showed a growing interest on executives' participation in crisis communication (Jin et al., 2018; Schoofs & Claeys, 2021; Zhang et al., 2024) and understanding online public opinion during crises (e.g., Ji & Kim, 2021; Rim et al., 2020; Xue et al., 2024). Page et al. (2023) underscored the need to dedicate more research on public perspectives of crises. Echoing this burgeoning focus on analyzing and responding to publics' concerns expressed on social media, this study adopting a public-centric approach examined the public opinion that occurred on social media in response to Zoom CEO's authentic communication of a layoff crisis. LDA-based topic modeling and VADER sentiment analysis were adopted to identify major topics and classify the polarity of expressed sentiment among posts collected from various social media platforms (i.e., X, YouTube, Facebook, and Reddit). As corporate reputation is intrinsically shaped by online public opinion (Kotras, 2020; Meadows & Meadows, 2016), this study provided empirical support for involving CEO and engaging in authentic communication to attenuate negative public opinion on social media during crisis. Findings of this study contribute to the existing scholarship on the role of leadership communication during crisis and the value of understanding online public opinion for reputation management.

This study yielded three key findings. First, online publics showed positive reactions toward the CEO's authentic communication of the layoff announcement. Specifically, the CEO's authentic communication (i.e., admitting mistakes, taking pay cut, displaying empathy for the affected employees, embracing the company's core values, etc.) of the layoffs seemed to have gained favorable public opinion (e.g., appreciation, praise, respect, admiration, etc.) from publics on social media. This positive reaction was displayed through the identified major topics linking these attributes in the CEO's layoff communication and the positive sentiment scores toward both the company and the CEO. Online publics' frequent mentions of some key phrases (e.g., "sorry," "taking personal responsibility," "making sacrifices," "taking a cut," "selfless act," "admitting mistakes," etc.) exemplifying authentic communication revealed that Zoom CEO's communication indeed led online publics to focus on these defining characteristics of authentic communication. Such focus fostered a positive sentiment both toward the company and its CEO. Second, the rising trend of positive sentiment over negative sentiment voiced on social media served as clear evidence for the

effectiveness of the CEO's authentic communication during this challenging time. Furthermore, the dominance of neutral sentiment during the entire observation period indirectly suggested that most publics on social media did not react negatively toward the organization and its CEO, despite this relatively larger magnitude of downsizing compared to previous downsizings in the tech industry. Third, the positive sentiment appeared to have transferred to Zoom's reputation as an organization. Such positive halo effect or contagion (Hingston et al., 2017) underscored the mitigation power of CEO's authentic communication to offset negative sentiment in a para-crisis. Given this prominent role of executives in crisis management, some communication scholars (Liu, Hong, & Yook, 2022; Schmitt, Xin, & Langan, 2020) explicitly addressed CEO as *Chief Crisis Officer*. These findings have significant theoretical implications for leadership communication in crisis management and the role of organizational executives on reputation management during crisis, which is explained in the following section.

5.1. Theoretical implications

Leadership communication in crisis. Findings of this study lend credence to the role of leadership communication in crisis management. Effective crisis communication can shift public opinion and temper negative crisis impacts (Coombs & Holladay, 2022; Liu et al., 2024). Jin et al. (2018) explicitly stressed that a formula for successful crisis communication from leaders should contain honesty, transparency, and integrity, which are key attributes of authentic communication. Given the shortage of research linking executives' authentic communication to crisis, this study advances the current body of scholarship on authentic communication, especially the topic of executives' authentic communication in crisis, in the following aspects. First, this study diversifies the existing approach to provide validation to the effectiveness of authentic crisis communication by organizational leaders. Most research on the effects of authentic communication has been conducted through surveys and experiments (Barkey & Okamoto, 2023; Lim & Jiang, 2021; Men, 2021; Men et al., 2018). Echoing public relations scholars' call to ground crisis examination in "public perspectives of crisis" (Page et al., 2023, p. 3), this study represents a growing scholarship in public relations that has examined the near real- time public reactions toward a CEO's authentic communication during crisis. Such approach provides timely information about the needs and concerns of online publics that are conducive to formulating crisis response strategies and assess the effectiveness of crisis communication.

Second, this study expands the limited literature on authentic communication by organizational leaders during a crisis. The CEO's display of full information disclosure, demonstrating care and empathy, and showing accountability in the layoff announcement seemed to mitigate the negative sentiment typically associated with mass layoffs. The association found in this study between Zoom CEO's authentic crisis communication (e.g., admitting mistakes, explaining reasons for layoff, and taking accountability) and positive sentiments expressed by online publics attests the value of authentic communication to positively affect the cognitive (i.e., thoughts) and emotional (i.e., sentiment) reactions (i.e., public opinion) from online publics when organizational leaders engage in crisis communication that is genuine, transparent, and consistent (Erickson, 2021; Liu et al., 2024; Men, 2021). Zoom's CEO seemed to have significantly benefited from his authentic layoff announcement as the sentiment surrounding the public discourse on the CEO was positive contrary to the expectation that publics would hold the CEO accountable for the mass layoff decision. Particularly, Zoom CEO's authentic communication during the layoff crisis created a buffering effect against the negative sentiment toward the company during the mass layoffs.

Third, this study illustrates that an underlying appeal of authentic communication by leaders during crisis lies in demonstrating shared values and care to stakeholders or online publics. This distinct appeal of authentic communication corresponds to a values and care-centered approach to crisis communication (McGuire et al., 2020), which focuses on highlighting shared values, creating common goals grounded in shared values, and prioritizing stakeholders' welfare. To reinforce Zoom's core value of "care" as prominently indicated on the company's Website, the CEO stressed in the layoff announcement that employees are what matters most to the company and outlined detailed plans on how to provide financial, psychological, and career support to those affected employees. Especially, the CEO's self-sacrifice (i.e., substantial cuts to salary and bonus) undoubtedly demonstrates the CEO's unwavering commitment to prioritize stakeholders' welfare before his own self-interest. The alignment between authentic communication and a values and care-centered approach to crisis communication provides further conceptual support for incorporating authentic communication in executives' crisis communication (e.g., Rees, 2020; Shirdastian et al., 2019; Zhang et al., 2019).

The Role of CEOs on corporate reputation management during crisis. Based on the results from topic modeling and sentiment analysis, the Zoom CEO was perceived as a genuine, accountable, and consistent leader by publics on social media. Such favorable perceptions seemed to have transferred to the target organization, resulting in positive attitudes for the company, Zoom. This finding can find a meaningful explanation from the meaning transfer model (McCracken, 1989) where symbolic meanings embodied by an organization's spokesperson or representative can be projected onto the target organization, hence fostering desirable perceptions such as brand reputation (e.g., Furukawa, 2022; Kim & Ji, 2021). Originally conceptualized as a mechanism to illustrate how certain subjective, symbolic meanings embedded in a brand endorser is projected onto a product or brand through celebrity endorsement (Schimmelpfennig & Hunt, n.d.). Similar to an endorser of a brand, a CEO acts as a symbolic spokesperson of an organization. The desirable attributes perceived from a CEO can thus be transferred to the focal organization as well. When publics on social media perceived that Zoom CEO, Eric Yuan, genuinely cared about the employees and acted selflessly to hold himself accountable for the company's mistakes, these positive attributes seemed to be transferred onto the focal organization, Zoom. It is crucial to point out that Zoom CEO's authentic communication during the layoff crisis played a pivotal role in making these desirable traits salient to online publics, thus making such meaning transfer possible. This perceptual process underscores the centrality for organizational executives to strive for strategic communication strategies such as authentic communication to allow their positive qualities to be transferred to the focal organization, hence protecting an organization's reputation in turbulent situations (e.g., para-crisis).

5.2. Practical implications

The findings of this study offer practical implications for both public relations practitioners and organizational executives in crisis management. The fundamental understanding and competence on how to utilize social media analytics tools to conduct social listening, analyze public opinion, and perform sentiment analysis is essential to analyzing crises in real-time. Understanding online public opinion in shaping reputation assessment is crucial. Thus, recruitment and training efforts need to be directed to cultivate internal (e.g., in-house practitioners) or external (e.g., contracted consultants) talents in social media analytics, capable of programming languages or utilization of various social media analytics tools and software (e.g., Brandwatch, Talkwalker, Meltwater, etc.). During a crisis, specialized public relations practitioners should be designated to actively monitor, analyze online public opinion, and communicate with online publics (e.g., responding to complaints or negative eWOM), especially tracking and assessing potential risks from negative online discourse to reputation damages. Additionally, given the significant role of organizational executives in shaping online public opinion during crises, CEOs or organizational leaders should be encouraged to proactively participate in crisis communication to

address the concerns and needs of publics based on the analysis of online public opinion in order to protect the focal organization's reputation. Lastly, executives should recognize the value of authentic communication and incorporate it in crisis management to project desirable attributes as crisis leaders who prioritize stakeholders' welfare, focus on shared values, and exhibit transparency, accountability, and consistency.

5.3. Limitations and future research directions

Limitations present themselves as in any study. One common limitation of research utilizing social media data is the challenge of generalization, as social media users may not accurately represent the broader population (Hargittai, 2020). However, analyzing real-time discussions offers significant value, particularly during crises when timely and well-informed responses are critical (Choi, 2021; McKinley et al., 2024; Zhang et al., 2024). As the data collected were primarily sourced from X, which may limit representativeness and introduce potential sampling bias, including an increased influence of outliers. Despite this, research has consistently demonstrated the value of social media data from platforms such as X (Shah & Schweiggart, 2023; Xue et al., 2024), YouTube (Choi & Park, 2024; Kwon & Park, 2023), and TikTok (Che & Kim, 2023; McDermott, 2024) in capturing real-time public reactions during crises. Future research could explore whether public opinion varies across different platforms, as each tends to attract distinct demographic groups.

Next, despite the extensive measures to refine the data collection process, some uncertainty inevitably remains regarding whether every individual post can be viewed as a direct response to the CEO's authentic layoff communication. An initial review of the data revealed that social media users reacted by quoting the CEO's layoff announcement, sharing a direct link to the announcement, or posting links to news articles referencing the CEO's announcement. To verify relevance, we randomly checked posts containing web or news links to ensure that they indeed led to the CEO's announcement or included direct quotes from that announcement. Nevertheless, manually verifying every linked post was not feasible, which prevented us from definitively confirming that all posts reflected online publics' reactions to the CEO's authentic layoff communication. Additionally, our search queries contained "employees" as a keyword in data collection, which might have led to misclassified or irrelevant data. Recognizing this, we took meticulous measures to remove any post containing "employees" that did not explicitly reference Zoom layoffs in both datasets. Furthermore, another limitation pertains to the potential overlap between the company and CEO-related datasets. As both search queries included the word "Zoom," posts from both datasets might reference "Zoom," thus resulting in overlapping data. While we made efforts to separate the two collections, this overlap is inevitable when using common terminology to search for both a company and its CEO. Such overlap could result in some data redundancy, which highlights the challenge of distinguishing the two entities in a public discourse context. Therefore, similarity scores were generated to document the number of overlapping posts. However, it should be noted that some overlapping posts were appropriately collected in both datasets as numerous posts discussed both the Zoom brand and its CEO. Finally, given the limited scope of this study on one organization's para-crisis related to a specific incident (i.e. layoff), it is challenging to adequately demonstrate the mitigating impact of authentic communication on the reputational threat to a brand and its CEO on social media during crises. Future research is thus encouraged to explore this linkage in more crisis contexts such as product recalls, scandals, or spillover crises.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Ables, K. (2023, February 9). *Zoom was a pandemic darling. Now it's part of the tech layoffs wave. Washington Post*. Retrieved from: ⟨https://www.washingtonpost.com/technology/2023/02/09/zoom-layoffs-ceo-tech-industry-covid/⟩.

Agarwal, S. (2019). Deep learning-based sentiment analysis: Establishing customer dimension as the lifeblood of business management. *Global Business Review, 23*, 119–136.

Anders, A. D. (2021). Human-centered leadership development: A communication-based approach for promoting authentic and transformational leadership. *International Journal of Business Communication, 62*(3), 608–631.

Atkins, R., & Favreau, C. (2022). The effects of layoffs and plant closings on manufacturers' market value. *International Journal of Production Economics, 245*, 108392.

Avery, E. J. (2017). Public information officers' social media monitoring during the Zika virus crisis, a global health threat surrounded by public uncertainty. *Public Relations Review, 43*, 468–476.

Baker, M. J., & Hashimoto, B. (2024). Expression of customer (dis)satisfaction in online restaurant reviews: The relationships between adversative connective constructions and star ratings. *International Journal of Business Communication, 61*(1), 148–180.

Barachi, E. M., Alkhatib, M., Mathew, S., & Oroumchian, F. (2021). A novel sentiment analysis framework for monitoring the evolving public opinion in real-time: Case study on climate. *Journal of Cleaner Production, 312*, Article 127820. https://doi.org/10.1016/j.jclepro.2021.127820

Barkey, K., & Okamoto, E. (2023). Breaking the mold: The influence of CEO gender and ethnicity on crisis communication effectiveness in Japan. *Asian Pacific Communication, 33*(1), 38–60.

Barkley, K. (2020). The impact of CEO ethnicity and language choice on crisis communication in Japan. *International Journal of Business Communication, 57*(2), 244–259.

Benlahbib, A., & Nfaoui, E. H. (2019). A hybrid approach for generating reputation based on opinions fusion and sentiment analysis. *Journal of Organizational Computing and Electronic Commerce, 30*(1), 9–27.

Benlahbib, A., & Nfaoui, E. H. (2020). Aggregating customer review attributes for online reputation generation. *IEEE Access, 8*, 96550–96564.

Bergh, D. D., Connelly, B. L., Ketchen, D. J., & Shannon, L. M. (2014). Signaling theory and equilibrium in strategic management research: An assessment and research agenda. *Journal of Management Studies, 51*, 1334–1360.

Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent dirichlet allocation. *Journal of Machine Learning Research, 3*, 993–1022.

Bonta, V., Kumaresh, N., & Janardhan, N. (2019). A comprehensive study on lexicon based approaches for sentiment analysis. *Asian Journal of Computer Science and Technology, 8*(2), 1–6.

Bowen, S. A. (2016). Clarifying ethics terms in public relations from A to V, authenticity to virtue. *Public Relations Review, 42*, 564–572.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology, 3*(2), 77–101.

Braun, V., & Clarke, V. (2022). Conceptual and design thinking for thematic analysis. *Qualitative Psychology, 9*(1), 3–26.

Brodsky, A.. January 20). Communicating authenticity in a virtual world. *Havard Business Review*. https://hbr.org/2022/01/communicating-authentically-in-a-virtual-world.

Brown-Devlin, N., Lim, H. S., & Tao, J. (2022). Examining the Influence of Algorithmic Message Personalization on Source Credibility and Reputation. *International Journal of Business Communication*, 1–24.

Byrd, K., & John, R. S. (2022). Lies, damned lies, and social media following extreme events. *Risk Analysis, 42*(8), 1704–1727.

Calissano, A., Vantini, S., & Arena, M. (2020). Monitoring rare categories in sentiment and opinion analysis: A Milan mega event on Twitter platform. *Statistical Methods Applications, 29*, 787–812.

Celik, A., Akgemci, T., & Akyazi, E. E. (2016). A comparison between the styles of transformational leaders and authentic leaders in crisis management. *International Journal of Academic Research in Business and Social Sciences, 6*(2), 195–208.

Chang, K., Yang, M., Zhou, S., & Wei, G. (2024). The impacts of online public opinions on stock price synchronicity in China: Evidence from stock forums. *Expert Systems with Applications, 255*, Article 124520 (Article).

Che, S. P., & Kim, J. H. (2023). Sentiment impact of public health agency communication strategies on TikTok under COVID-19 normalization: Deep learning exploration. *Journal of Public Health*. https://doi.org/10.1007/s10389-023-01921-5

Chen, K., & Tomblin, D. (2021). Using data from reddit, public deliberation, and surveys to measure public opinion about autonomous vehicles. *Public Opinion Quarterly, 85*, 289–322. https://doi.org/10.1093/poq/nfab021

Chen, X., Zhang, W., Xu, X., & Cao, W. (2022). A public and large-scale expert information fusion method and its application: Mining public opinion via sentiment analysis and measure public dynamic reliability. *Information Fusion, 78*, 71–85.

Cheng, Y., Huang, Y.-H., C, & Chan, C. M. (2017). Public relations, media coverage, and public opinion in contemporary China: Testing agenda building theory in a social mediated crisis. *Telematics and Informatics, 34*(3), 765–773.
Cho, M., Xu, S., & Boatwright, B. (2023). A personal-communicative evaluation approach to CEO advocacy & employee relations. *Public Relations Review, 49*, Article 102295.
Choi, J. A. (2021). Pro-maskers vs anti-maskers during COVID-19: The role of social media in fostering social networking and detecting reaction to prevention recommendations between U.S. and South Korea. *Društvena istraživanja. Journal for General Social Issues, 30*(2), 359–378.
Choi, J. A., & Park, S. (2024). Examining public responses to transgressions of CEOs on YouTube: Social and semantic network analysis. *Journal of Contemporary Eastern Asia, 23*(10), 18–34.
Cinelli, M., Morales, G. D. F., Galeazzi, A., Quattrociocchi, W., & Starnini, M. (2021). The echo chamber effect on social media. *Proceedings of the National Academy of Sciences of the United States of America, 118*(9). https://doi.org/10.1073/pnas.2023301118
Claeys, A.-S., & Waele, A. D. (2022). The importance of authenticity in organizational crisis communication via social media. In Y. Jin, & L. Austin (Eds.), *Social media and crisis communication* (2nd ed., pp. 106–116). New York: Routledge.
Clementson, D. E. (2020). Narrative persuasion, identification, attitudes, and trustworthiness in crisis communication. *Public Relations Review, 46*(2), Article 101889 (Article).
Coombs, W. T., & Holladay, J. S. (2012). The paracrisis: The challenges created by publicly managing crisis prevention. *Public Relations Review, 38*, 408–415.
Coombs, W. T., & Holladay, J. S. (2014). How publics react to crisis communication efforts: Comparing crisis response reactions across sub-arenas. *Journal of Communication Management, 18*(1), 40–57.
Coombs, W. T., & Holladay, J. S. (2022). Crisis communication as strategic communication: Process and insights. In J. Falkheimer, & M. Heide (Eds.), *Research handbook on strategic communication* (pp. 259–273). Cheltenham, UK: Edward Elgar.
Dahal, B., Kumar, S. A., & Li, Z. (2019). Topic modeling and sentiment analysis of global climate change tweets. *Social Network Analysis and Mining, 9*, 1–20.
Dai, J., Zhao, Y., & Li, Z. (2024). Sentiment-topic dynamic collaborative analysis-based public opinion mapping in aviation disaster management: A case study of the MU5735 air crash. *International Journal of Disaster Risk Reduction, 102*, Article 104268. https://doi.org/10.1016/j.ijdrr.2024.104268
Dailey, M. (2021). *A leader's guide to storytelling: Restoring authentic communication in a world of change*. Routledge.
Datta, D. K., Guthrie, J., Basuil, D., & Pande, A. (2010). Causes and effects of employee downsizing: A review and synthesis. *Journal of Management, 36*(1), 281–348.
Deephouse, D. L., Newburry, W., & Soleimani, A. (2016). The effects of institutional development and natural culture on cross-national differences in corporate reputation. *Journal of World Business, 51*(3), 463–473.
Deng, Q., Gao, Y., Wang, C., & Zhang, H. (2020). Detecting information requirements for crisis communication from social media data: An interactive topic modeling approach. *International Journal of Disaster Risk Reduction, 50*, Article 101692 (Article).
Denner, N., Koch, T., & Senger, S. (2021). Faces of companies: Personalization of corporate coverage in crisis and non-crisis periods. *Journalism, 22*(11), 2747–2763.
Denner, N., Viererbl, B., & Koch, T. (2019). A matter for the boss? How personalized communication affects recipients' perceptions of an organization during a crisis. *International Journal of Communication, 13*, 2026–2044.
Diesner, J., & Carley, K. M. (2010). Methodology for integrating network theory and topic modeling and its application to innovation diffusion. *2010 IEEE Second International Conference on Social Computing* (pp. 687–692). IEEE (, August).
Dong, C., Harrison, V., & Zheng, Q. (2024). When controversy meets controversy, what do you see? Understanding visual communication of cannabis companies' corporate social advocacy. *Public Relations Review, 50*, 102417.
Dong, Y., Li, Y., Hua, H.-Y., & Li, Wei (2023). Perceived tourism authenticity on social media: The consistency of ethnic destination endorsers. *Tourism Management Perspectives, 49*, Article 101176.
Edwards-Levy, A. (2021). *Here's what pollsters think happened with 2020 election surveys. CNN*. Retrieved from: ⟨https://www.cnn.com/2021/05/13/politics/2020-polling-error-research/index.html?utm_term⟩¼link&utm_source¼twCNN&utm_medium¼social&utm_content¼2021-05-14T12%3A52%3A28.
Einwiller, S., & Steilen, S. (2015). Handling complaints on social network sites: An analysis of complaints and complaint responses on Facebook and Twitter pages of large US companies. *Public Relations Review, 41*(2), 195–204.
Elbagir, S., & Yang, J. (2019). Twitter sentiment analysis using natural language toolkit and VADER sentiment (, March) *In Proceedings of the International Multiconference Of Engineers And Computer Scientists, 122*, 16.
Erickson, S. (2021). Communication in a crisis and the importance of authenticity and transparency. *Journal of Library Administration, 61*(4), 476–483.
Fan, C., Zhang, C., Yahja, A., & Mostafavi, A. (2021). Disaster city digital twin: A vision for integrating artificial and human intelligence for disaster management. *International Journal of Information Management, 56*, Article 102049.
Fan, X.-L., Zhou, Y., Wang, C.-X., & Chang, X.-X. (2020). The interactive effect of employee- involved governance and CEO change-oriented leadership on organizational innovation: A moderated mediation model. *Group Organization Management, 45*(3), 417–455.
Fombrun, C. J. (2005). Building corporate reputation through CSR initiatives: Evolving standards. *Corporate Reputation Review, 8*, 7–11.
Ford, B., & Bloomberg. (2023, February 7). Zoom is the latest big tech company to announce major layoffs at it cuts 1,300 jobs after tripling workforce. Fortune. Retrieved at ⟨https://fortune.com/2023/02/07/zoom-layoffs-1300-jobs-cut-pandemic-hiring-spree/⟩.
Fox, C., Davis, P., & Baucus, M. (2021). Corporate social responsibility during unprecedented crises: The role of authentic leadership and business model flexibility. *Management Decision, 58*(10), 2213–2233.
Furukawa, H. (2022). How does the CEO's influence affect consumer brand trust? The mediating effects of symbolic and environmental product perceptions. *Journal of International Consumer Marketing, 34*(1), 11–23.
Gao, W., Zheng, C., Zhu, X., Deng, H., Wang, Y., & Hu, G. (2024). Knowledge-injected prompt learning for actionable information extraction from crisis-related tweets. *Computers and Electrical Engineering, 118*. Article 109398.
Grunig, J. E. (1992). Communication, public relations, and effective organizations: An overview of the book. In J. E. Grunig (Ed.), *Excellence in public relations and communication management* (pp. 1–28). Mahwah, NJ: Laurence Erlbaum.
Guèvremont, A., & Grohmann, B. (2018). Does brand authenticity alleviate the effect of brand scandals? *Journal of Brand Management, 25*, 322–336.
Hambrick, D. C., & Wowak, A. J. (2021). CEO sociopolitical activism: A stakeholder alignment model. *The Academy of Management Review, 46*(1), 33–59.
Hargittai, E. (2020). Potential biases in big data: Omitted voices on social media. *Social Science Computer Review, 38*(1), 10–24.
Harrison, V., Vafeiadis, M., Diddi, P., & Conlin, J. (2022). The impact of CSR on nonprofit outcomes: How the choice of corporate partner influences reputation and supportive intentions. *Corporate Communications: An International Journal, 27*(2), 205–225.
Hayes, A. F., & Krippendorff, K. (2007). Answering the call for a standard reliability measure for coding data. *Communication Methods and Measures, 1*(1), 77–89.
Hearit, K. M. (2021). A blame narrative approach to apologetic crisis management: The serial apologiae of United Airlines. *Public Relations Review, 47*(5), Article 102106.
Hellerstein, J. K., Kutzbach, M. J., & Neumark, D. (2019). Labor market networks and recovery from mass layoffs: Evidence from the Great Recession period. *Journal of Urban Economics, 113*, 103192.
Hennig-Thurau, T., Gwinner, K. P., Walsh, G., & Gremler, D. D. (2004). Electronic word-of- mouth via consumer-opinion platforms: What motivates consumers to articulate themselves on the Internet? *Journal of International Marketing, 18*(1), 38–52.
Hickman, L., Thapa, S., Tay, L., Cao, M., & Srinivasan, P. (2022). Text preprocessing for text mining in organizational research: Review and recommendations. *Organizational Research Methods, 25*(1), 114–146.
Hingston, S. T., McManus, J. F., & Noseworthy, T. J. (2017). How inferred contagion biases dispositional judgements of others. *Journal of Consumer Psychology, 27*(2), 195–206.
Hong, S., & Cameron, G. T. (2018). Will comments change your opinion? The persuasion effects of online comments and heuristic cues in crisis communication. *Journal of Contingencies and Crisis Management, 26*, 173–182.
Hoti, M. H., & Ajdari, J. (2023). Sentiment analysis using the VADER model for assessing company services based on posts on social media. *SEEU Review, 18*(2), 19–33.
Huang, M., & Ki, E.-J. (2023). Protecting organizational reputation during a para-crisis: The effectiveness of conversational human voice on social media and the roles of construal level, social presence, and organizational listening. *Public Relations Review, 49*, Article 102389.
Hutto, C., & Gilbert, E. (2014). *VADER: A parsimonious rule-based model for sentiment analysis of social media text. Proceedings of the International AAAI Conference on Web and Social Media*, 216-225, Ann Arbor: MI.
Jabalameli, S., Xu, Y., & Shetty, S. (2022). Spatial and sentiment analysis of public opinion toward COVID-19 pandemic using twitter data: At the early stage of vaccination. *International Journal of Disaster Risk Reduction, 80*, Article 103204. https://doi.org/10.1016/j.ijdrr.2022.103204
Ji, Y. G., & Hong, C. (2023). Engaging employees in CEO activism: The role of transparent leadership communication in making a social impact. *Journalism & Mass Communication Quarterly, 100*(1), 78–99.
Ji, Y., & Kim, S. (2021). Public expectations of crisis outcomes in the social media era in China: A communication-mediated psychological mechanism. *Public Relations Review, 47*(5), Article 102119.
Jiang, H., & Luo, Y. (2018). Crafting employee trust: From authenticity, transparency to engagement. *Journal of Communication Management*, 138–160.
Jin, J., Austin, L., Eaddy, L., Spector, S., Reber, B., & Espina, C. (2018). How financial crisis history informs ethical corporate communication: Insights from corporate communication leaders. *Public Relations Review, 44*, 574–584.
Jin, J., Mitson, R., Qin, Y. S., Vielledent, M., & Men, L. R. (2023). Enhancing young consumer's relational and behavioral outcomes: The impact of CEO activism authenticity and value alignment. *Public Relations Review, 49*, Article 102312.
Jungherr, A., Schoen, H., Posegga, O., & Jürgens, P. (2017). Digital trace data in the study of public opinion: An indicator of attention toward politics rather than political support. *Social Science Computer Review, 35*(3), 336–356.
Kelly, J.. *Zoom CEO Takes A Pay Cut After Laying Off 1,300 Employees And Sends Stock Price Soaring—Earning Him More Money. Forbes*. https://www.forbes.com/sites/jackkelly/2023/02/08/zoom-ceo-takes-a-pay-cut-after-laying-off-1300-employees-and-sends-stock-price-soaring-earning-him-more-money/?sh=4c712c2e38b4.
Kernis, M. H., & Goldman, B. M. (2006). A multicomponent conceptualization of authenticity: Theory and research. In M. P. Zanna (Ed.), *Advances in experimental social psychology, 38* pp. 283–357). Elsevier Academic Press.
Kim, S., & Ji, Y. (2021). Positive ripple effects of corporate leaders' CSR donations amid COVID-19 on corporate and country reputations: Multi-level reputational benefits of CSR focusing on Bill Gates and Jack Ma. *Public Relations Review, 47*(4), Article 102073.
Kim, S., & Krishna, A. (2018). Unpacking public sentiment toward the government: How citizens' perceptions of government communication strategies impact public engagement, cynicism, and communication behaviors in South Korea. *International Journal of Strategic Communication, 12*(3), 215–236.

Kim, S., Krishna, A., & Plowman, D. (2018). Winning in the court of public opinion: Exploring public relations-legal collaboration during organizational crisis. *Corporate Communications: An International Journal, 24*(1), 96–114.
Kim, S., Zhang, X. A., & Zhang, B. W. (2016). Self-mocking crisis strategy on social media: Focusing on Alibaba chairman Jack Ma in China. *Public Relations Review, 42*(5), 903–912.
Kinsky, E. S., Chen, L., & Drumheller, K. (2021). Crisis and emergency risk communication: FEMA's Twitter use during the 2017 hurricane season. *Public Relations Review, 47*(4), Article 102094.
Koch, T., Denner, N., & Coutandin, F. (2022). A CEO but also a parent: How strategic communication of private information about the CEO affects perceptions of an organization during a crisis. *International Journal of Strategic Communication, 16*(4), 620–632.
König, A., Stöcklein, B., Hiller, N. J., Cooper, C. D., & Bong, D. (2025). Good fun or laughingstock? How CEO humor affects infomediaries' social evaluations of organizations. *Academy of Management Review, 50*(3), 581–611.
Kotras, B. (2020). Opinions that matter: The hybridization of opinion and reputation measurement in social media listening software. *Media, Culture Society, 42*(7-8), 1495–1511.
Krishna, A., & Kim, S. (2020). Exploring customers' situational and word-of-mouth motivations in corporate misconduct. *Public Relations Review, 46*(2), Article 101892.
Kumar, R., Pandey, D. K., & Goodell, J. W. (2023). Market reactions to layoff announcements during crises: Examining impacts and conditioners. *Finance Research Letters, 58*, 104423.
Kwon, S., & Park, A. (2023). Examining thematic and emotional differences across Twitter, Reddit, and YouTube: The case of COVID-19 vaccine side effects. *Computers in Human Behavior, 144*, Article 107734.
Lang, K., & Lang, G. E. (2009). Mass society, mass culture, and mass communication: The meanings of mass. *International Journal of Communication, 3*, 998–1024.
Laufer, D., Garrett, T. C., & Ning, B. (2018). The moderating role of power distance on the reaction of consumers to the CEO as a spokesperson during a product harm crisis: Insights from China and South Korea. *Journal of International Management, 24*(3), 215–221.
Lee, Y., & Dong, E. (2023). How transparent internal communication from CEO, supervisors, and peers leads to employee advocacy. *Management Communication Quarterly, 37*(4), 878–912.
Lee, Y., & Kim, K. H. (2020). De-motivating employees' negative communication behaviors on anonymous social media: The role of public relations. *Public Relations Review, 46*(4), Article 101955.
Lee, Y., & Tao, W. (2021). Does perceived morality of CEO activism matter? Understanding employees' responses to CEO actions on sociopolitical issues. *Management Decision, 59*(10), 2329–2354.
Li, C., & Ken, M. L. (2021). Explorations on mediated communication and beyond: Toward a theory of social media. *Public Relations Review, 47*, Article 102112.
Lim, J. S., & Jiang, H. (2021). Linking authenticity in CSR communication to organization- public relationship outcomes: Integrating theories of impression management and relationship management. *Journal of Public Relations Research, 33*(6), 464–486.
Lim, J. S., & Young, C. (2021). Effects of issue ownership, perceived fit, and authenticity in corporate social advocacy on corporate reputation. *Public Relations Review, 47*, Article 102071.
Lin, Y. R., Margolin, D., Keegan, B., & Lazer, D. (2013). Voices of victory: A computational focus group framework for tracking opinion shift in real time. *In Proceedings of the 22nd International Conference on World wide web*, 737–748. https://doi.org/10.1145/2488388.2488453
Liu, B. (2020). *Sentiment analysis: Mining opinions, sentiments, and emotions*. Cambridge University Press.
Liu, H., Cutcher, L., & Grant, D. (2017). Authentic leadership in context: An analysis of banking CEO narratives during the global financial crisis. *Human Relations, 70*(6), 694–724.
Liu, J., Hong, C., & Yook, B. (2022). CEO as "Chief Crisis Officer" under COVID-19: A content analysis of CEO open letters using structural topic modeling. *International Journal of Strategic Communication, 16*(3), 444–468.
Liu, A., Shu, C., & Xiao, Z. (2024). Entrepreneurial orientation, political ties, and corporate reputation: The moderating roles of institutional environment. *Journal of Business Research, 170*, Article 114347.
Lock, O., & Pettit, C. (2020). Social media as passive geo-participation in transportation planning—How effective are topic modeling & sentiment analysis in comparison with citizen surveys? *Geo-Spatial Information Science. 23*, 275–292.
Lopes Junior, D. da S., Matos, J. S. A. de, Cardoso, T. G., & Zukowsky-Tavares, C. (2024). The authentic leaders in times of crises: A portrait of the effects of authentic leadership on the well-being of its employees during the COVID-19 pandemic. International Journal of Scientific Management and Tourism, 10(2), e826. https://doi.org/10.55905/ijsmtv10n2-045.
Lu, X., & Jin, Y. (2020). Information vetting as a key component in social-mediated crisis communication: An exploratory study to examine the initial conceptualization. *Public Relations Review, 46*, Article 101891.
Ma, J., Sachdev, A. R., & Gu, X. (2020). Being oneself and doing great: The effect of authenticity on job performance and the role of supportive leadership. *Journal of Personnel Psychology, 19*(2), 75–85.
Manika, D., Papagiannidis, S., & bourlakis, M. (2017). Understanding the effects of a social media service failure apology: A comparative study of customers vs. potential customers. *International Journal of Information Management, 37*(3), 214–228.
McClimon, T. J. (2023, September). 25). Employees want more authentic communications from their leaders. Forbes. https://www.forbes.com/sites/timothyjmcclimon/2023/09/25/employees-want-more-authentic-communications-from-their-leaders/.
McCracken, G. (1989). Who is the celebrity endorser? Cultural foundations of the endorsement process. *Journal of Consumer Research, 16*, 310–321.
McDermott, V. (2024). Framing, agency, and athletic activism: The case of Simone Biles at the 2020 Olympics. *Public Relations Review, 50*, Article 102457.
McGregor, J.. *Forget Zoom layoffs. Job cuts by email are tech's latest digital pink slip. Forbes*. https://www.forbes.com/sites/jenamcgregor/2023/02/01/forget-zoom-layoffs-job-cuts-by-email-are-techs-latest-digital-pink-slip/.
McGuire, D., Cunningham, J. E. A., Reynolds, K., & Matthews-Smith, G. (2020). Beating the virus: An examination of the crisis communication approach taken by New Zealand Prime Minister Jacinda Ardern during the Covid-19 pandemic. *Human Resource Development International, 23*(4), 361–379.
McKinley, C. J., Choi, J. A., & Luo, Y. (2024). Expressing uncertainty and risk about the Mpox outbreak: A textual analysis of Twitter messaging. *Communication Studies, 75*(3), 283–301.
Meadows, C. Z., & Meadows, C. W. (2016). Corporate Reputation on Mass Media, Public Opinion, and YouTube: Examining the Factors Influencing Reputation of Fortune 500 Companies. *Corporate Reputation Review, 19*, 345–356.
Men, L. R. (2015). The internal communication role of the chief executive officer: Communication channels, style, and effectiveness. *Public Relations Review, 41*(4), 461–471.
Men, L. R. (2021). The impact of startup CEO communication on employee relational and behavioral outcomes: Responsiveness, assertiveness, and authenticity. *Public Relations Review, 47*, Article 102078.
Men, L. R., Tsai, W-H. S., Chen, Z. F., & Ji, Y. G. (2018). Social presence and digital dialogic communication: Engagement lessons from top social CEOs. *Journal of Public Relations Research, 30*(3), 83–99.
Men, L. R., Ji, Y. G., & Chen, Z. F. (2020). Strategic communication for startups and entrepreneurs in China. *Routledge*.
Men, L. R., & Tsai, W.-H. (2016). Public engagement with CEOs on social media: Motivations and relational outcomes. *Public Relations Review, 42*, 932–942.
Mishra, M. S., & Sharma, R. W. (2019). Brand crisis-sentiment analysis of user-generated comments about @Maggi on Facebook. *Corporate Reputation Review, 22*, 48–60.
Monehin, D., & Diers-Lawson, A. (2022). Pragmatic optimism, crisis leadership, and contingency theory: A view from the C-suite. *Public Relations Review, 48*, Article 102224.
Morgan, A., & Wilk, V. (2021). Social media users' crisis response: A lexical exploration of social media content in an international sport crisis. *Public Relations Review, 47*, Article 102057.
Myint, P. Y. W., Lo, S. L., & Zhang, Y. (2024). Unveiling the dynamics of crisis events: Sentiment and emotion analysis via multi-task learning with attention mechanism and subject-based intent prediction. *Information Processing & Management, 61*(4), 103695.
Nayani, R., Baric, M., Patey, J., Fitzhugh, H., Watson, D., Tregaskis, O., & Daniels, K. (2022). Authenticity in the pursuit of mutuality during crisis. *British Journal of Management, 33*(3), 1144–1162.
Ndone, J., Warner, B., & Duffy, M. E. (2022). Emotional crisis communication: The effects of CEO's expression of guilt and anger on organizational reputation. *International Journal of Strategic Communication, 16*(5), 685–699.
Nelson, K. K. (2020). Computational grounded theory: A methodological framework. *Sociological Methods Research, 49*(1), 3–42.
Nokkaew, M., Nongpong, K., Yeophantong, T., Ploykitikoon, P., Arjharn, W., Siritaratiwat, A., Narkglom, S., Wongsinlatam, W., Remsungnen, T., Namvong, A., & Surawanitkun, C. (2023). Analyzing online public opinion on Thailand-China high-speed train and Laos-China railway mega-projects using advanced machine learning for sentiment analysis. *Social Network Analysis and Mining, 14*, 15. https://doi.org/10.1007/s13278-023-01168-8
Noor, N., Okhai, R., Jamal, T. B., Kapucu, N., Ge, Y. G., & Hasan, S. (2024). Social-media-based crisis communication: Assessing the engagement of local agencies in Twitter during Hurricane Irma. *International Journal of Information Management Data Insights, 4*, Article 100236.
Nowell, L., Norris, J. M., White, D. E., & Moules, N. (2017). Thematic analysis: Striving to meet the trustworthiness criteria. *International Journal of Qualitative Methods, 16*(1), 1–13.
Nugroho, R., Paris, C., Nepal, S., Yang, J., & Zhao, W. (2020). A survey of recent methods on deriving topics from Twitter: Algorithm to evaluation. *Knowledge and Information Systems, 62*, 2485–2519.
Oh, J., & Ki, E.-J. (2019). Factors affecting social presence and word-of-mouth in corporate social responsibility communication: Tone of voice, message framing, and online medium type. *Public Relations Review, 45*(2), 319–331.
Ostrovsky-Berman, E. (2021). Can CEOs change the market perception by giving interviews to the media? *Journal of Behavioral Finance, 22*, 65–73.
Page, T. G., Zhou, A., & Capizzo, L. W. (2023). Finding the path beyond reputation repair: A structural topic modeling analysis of the crisis communication paradigm in public relations. *Public Relations Review, 49*, Article 102349 (Article).
Papadopoulou, C., Vardarsuyu, M., & Oghazi, P. (2023). Examining the relationships between brand authenticity, perceived value, and brand forgiveness: The role of cross-cultural happiness. *Journal of Business Research, 167*, Article 114154.
Park, S., & Choi, J. A. (2023). Comparing public responses to apologies: Examining crisis communication strategies using network analysis and topic modeling. *Quality Quantity, 57*(4), 3603–3620.
Perkins, K. M. (2023). May 27). Authenticity: The key to great leadership and how to embrace it. Forbes. https://www.forbes.com/sites/kathymillerperkins/2023/05/27/authenticity-the-key-to-great-leadership-and-how-to-embrace-it/.

Portmann, E., Meier, A., Cudré-Mauroux, P., & Pedrycz, W. (2015). FOR A – A fuzzy set based framework for online reputation management. *Fuzzy Sets and Systems, 269* (15), 90–114.

Prichard, J., Watters, P., Krone, T., Spiranovic, C., & Cockburn, H. (2018). Social media sentiment analysis: A new empirical tool for assessing public opinion on crime? *Current Issues in Criminal Justice, 27*(2), 217–236.

Puschmann, C., & Powell, A. (2018). Turning words into consumer preferences: How sentiment analysis is framed in research and the news media. *Social Media and Society, 4*(3), Article 2056305118797724.

Qin, Y. S., DiStaso, M. W., Fitzsimmons, A. B., & Heffron, E. (2023). Communicating the big picture with employees: The impacts of CEO vision communication on employee engagement. *International Journal of Business Communication. Advance Online Publications.* https://doi.org/10.1177/23294884231190.

Qu, J. G., Yang, C. Y., Chen, A. A., & Kim, S. (2024). Collective empowerment and connective outcry: What legitimize netizens to engage in negative word-of-mouth of online firestorm? *Public Relations Review, 50*, Article 102438.

Rees, S. (2020). *Public Relations, branding and authenticity: Brand communications in the digital age*. New York: Routledge.

Reyes-Menendez, A., Saura, J. R., & Alvarez-Alonso, C. (2018). Understanding #WorldEnvronmentDay user opinions in Twitter: A topic-based sentiment analysis approach. *Sustainability. 15*(11), 2537. https://doi.org/10.3390/ijerph15112537

Rim, H., Lee, Y., & Yoo, S. (2020). Polarized public opinion responding to corporate social advocacy: Social network analysis of boycotters and advocators. *Public Relations Review, 46*(2), Article 101869. ⟨https://doiorg.ezproxy.lib.vt.edu/10.1016/j.pubrev.2019.101869⟩.

Rodrigues, P., Sousa, A., Lopes, J., & Borges, A. P. (2024). How does brand authenticity affect brand hate-love relationships and buying intention? The moderating role of brand personality. *Journal of Brand Strategy, 12*(4), 381–403.

Saura, J. R., Palacios-Marqués, D., & Ribeiro-Soriano, D. (2023). Exploring the boundaries of open innovation: Evidence from social media mining. Technovation. *Article, 119*, Article 102447.

Sauter, S., & Jungblut, M. (2024). It's good for our reputation (?!) – the impact of socio-political CEO communication on corporate reputation. *International Journal of Strategic Communication, 18*(1), 19–37.

Schmitt, A., Xin, K., & Langan, R. (2020). *Research: 3 biases that shaped CEO's pandemic response. Harvard Business Review, .⟨https://store.hbr.org/product/research-3-biases-that-shaped-ceos-pandemic-response/H05PWS⟩.*.

Schimmelpfennig, C., & Hunt, J.B. (n.d.)*Fifty years of celebrity endorser research: Support for a comprehensive celebrity endorsement strategy framework. Psychology & Marketing, 37*(3), 488-505.

Schoofs, L., & Claeys, A.-S. (2021). Communicating sadness: The impact of emotional crisis communication on the organizational post-crisis reputation. *Journal of Business Research, 130*, 271–282.

Schulz, A.-C., & Johann, S. (2018). Downsizing and the fragility of corporate reputation: An analysis of the impact of contextual factors. *Scandinavian Journal of Management, 34*(1), 40–50.

Scotland, J., Thomas, A., & Jing, M. (2024). *Public emotion and response immediately following the death of George Floyd: A sentiment analysis of social media comments. Telematics and Informatics Reports, 14*, 100143.

Shah, A. M., & Schweiggart, N. (2023). #BoycottMurree campaign on twitter: Monitoring public response to the negative destination events during a crisis. *International Journal of Disaster Risk Reduction, 92*(15), Article 103734.

Shamir, B., & Eilam-Shamir, G. (2018). What's your story?" A life-stories approach to authentic leadership development. In I. Katz, G. Eilam-Shamir, R. Kark, & Y. Berson (Eds.), *Leadership now: Reflections on the legacy of Boas Shamir* (pp. 51–76). UK: Emerald Publishing.

Shen, H., & Kim, J.-N. (2012). The authentic enterprise: Another buzz word, or a true driver of quality relationships? *Journal of Public Relations Research, 24*(4), 371–389.

Shirdastian, H., Laroche, M., & Richard, M.-O. (2019). Using big data analytics to study brand authenticity sentiments: The case of Starbucks on Twitter. *International Journal of Information Management, 48*, 291–307.

Silva, V. H., Duarte, A. P., & Oliveira, J. P. (2023). How does authentic leadership boost work engagement? Exploring the mediating role of work meaningfulness and work-family enrichment. *Administrative Sciences, 13*(10), 219–236.

Singh, A., Jenamani, M., Thakkar, J. J., & Rana, N. P. (2022). Quantifying the effect of eWOM embedded consumer perceptions on sales: An integrated aspect-level sentiment analysis and panel data modeling approach. *Journal of Business Research, 138*, 52–64.

Sisson, D. C., & Bowen, S. A. (2017). Reputation management and authenticity: A case study of Starbucks' UK tax crisis and "#SpreadTheCheer" campaign. *Journal of Communication Management, 21*(3), 287–302.

Södergren, J. (2021). Brand authenticity: 25 years of research. *International Journal of Consumer Studies, 45*(4), 645–663.

Spence, M. (1974). Competitive and optimal responses to signals: An analysis of efficiency and distribution. *Journal of Economic Theory, 7*(3), 296–332.

Stäbler, S., Himme, A., Edeling, A., & Backhaus, M. (2023). How firm communication affects the impact of layoff announcements on brand strength over time. *International Journal of Research in Marketing, 40*(3), 700–723.

Su, L., Stepchenkova, S., & Kirilenko, A. P. (2019). Online public response to a service failture incident: Implications for crisis communications. *Tourism Management, 73*, 1–12.

Sucher, S.J., & Westner, M.M. (2023). *Layoffs in the Tech industry: 2022-2023. Harvard Business School Case 323-095*. Retrieved at ⟨https://www.hbs.edu/faculty/Pages/item.aspx?num=63824⟩.

Tavazoee, F., Conversano, C., & Mola, F. (2020). Recurrent random forest for the assessment of popularity in social media. *Knowledge and Information Systems, 62*, 1847–1879.

Thelen, P. D., & Men, L. R. (2018). Strategic use of Facebook for public engagement in higher education institution. *Public Relations Journal, 12*(2), 1–27.

Thomsen, S. R. (2023). "Not the company we thought it was." Southwest Airlines' attempt at image repair during its October 2021 flight cancellation crisis. *Public Relations Review, 49*(2), Article 102309.

Tranfield, D., Denyer, D., & Smart, P. (2003). Towards a methodology for developing evidence- informed management knowledge by means of systematic review. *British Journal of Management, 14*, 207–222.

Tsai, W-H. S., & Men, L. R. (2017). Social CEOs: The effects of CEOs' communication styles and parasocial interaction on social networking sites. *New Media & Society, 19* (11), 1848–1867.

Ugoalah, C. (2015). Quality visibility in CEO response: Connecting to authenticity following a crisis. , October 7 - 10 *Paper presented at the 4th International Conference on Crisis Communication in the 21st Century Helsingborg, Sweden*.

Veena, G., Vinayak, A., & Nair, A. J. (2021). Sentiment analysis using improved Vader and dependency parsing. *Proceedings of the Global Conference for Advancement in Technology*, India, 1-6, doi: 10.1109/GCAT52182.2021.9587829.

Waddell, T. F. (2020). The authentic (and angry) audience. *Digital Journalism, 8*(2), 249–266.

Walther, J. B., & Whitty, M. T. (2021). Language, psychology, and new new media: The hyperpersonal model of mediated communication at twenty-five years. *Journal of Language and Social Psychology, 40*(1), 120–135.

Walumbwa, F., Avolio, B., Gardner, W., Wernsing, T., & Peterson, S. (2008). Authentic leadership: Development and validation of a theory-based measure. *Journal of Management, 34*(1), 89–126.

Wang, Y., Huang, Y.-H., & Cai, Q. (2022). Exploring the mediating role of government-public relationships during COVID-19 pandemic: A model comparison approach. *Public Relations Review, 48*, Article 102231.

Wang, Y., Wang, X., Zhang, J., Shi, M., & Wanta, W. (2024). Tracking attention about COVID-19 vaccines on twitter and newspapers: A dynamic agenda-setting approach. *Telematics and Informatics Reports, 13*, Article 100122 (Article).

Wang, X., & Zhang, C. (2024). Constructing the digital ark: How China utilizes shared media to foster online information collaboration and crisis management in public emergency response. *International Journal of Disaster Risk Reduction, 101*, Article 104239.

Weiss, M., & Muegge, S. (2019). Conceptualizing a new domain using topic modeling and concept mapping: A case study of managed security services for small businesses. *Technology Innovation Management Review, 9*(8), 55–64.

Westfall, C. (2018). *Leadership language: Using authentic communication to drive results*. Wiley.

Westrupp, E. M., Greenwood, C. J., Fuller-Tyszkiewicz, M., Berkowitz, T. S., Hagg, L., & Youssef, G. (2022). Texting mining of Reddit posts: Using Latent Dirichlet allocation to identify common parenting issues. *PLoS One, 17*(2), Article e0262529.

Wu, T., Reynolds, J., Wu, J., & Schlegelmilch, B. B. (2022). CEOs as corporate ambassadors: Deciphering leadership communication via Twitter. *Online Information Review, 46*(4), 787–806.

Xiao, Y., Hudders, L., Claeys, A., & Cauberghe, V. (2018). The impact of expressing mixed valence emotions in organizational crisis communication on consumers' negative word- of-mouth intention. *Public Relations Review, 44*(4), 794–806.

Xie, C., Zhang, J., Huang, Q., Chen, Y., & Morrison, A. M. (2022). An analysis of user-generated crisis frames: Online public responses to a tourism crisis. *Tourism Management Perspectives, 41*, Article 100931.

Xue, J., Guo, D., Zhao, Z., & Liu, S. (2024). How social media expedites the crisis spillover effect: A case study of Tesla's recall event. *Public Relations Review, 50*(1), Article 102432.

Yan, Z., Jing, X., & Pedrycz, W. (2017). Fusing and mining opinions for reputation generation. *Information Fusion, 36*, 172–184.

Yarchi, M., Baden, C., & Kligler-Vilenchik, N. (2021). Political polarization on the digital sphere: A cross-platform, over-time analysis of interactional, positional, and affective polarization on social media. *Political Communication, 38*(1–2), 98–139. https://doi.org/10.1080/10584609.2020.1785067

Yim, M. C. (2019). CEO's political tweets and perceived authenticity: Can expectancy violation be a pleasant surprise? *Public Relations Review, 45*(3), Article 101785.

Yu, D., Fang, A., & Xu, Z. (2023). Topic research in fuzzy domain: Based on LDA topic modeling. *Information Sciences, 648*, Article 119600 (Article).

Yue, C. A., Men, L. R., & Hart, E. B. (2023). Chief engagement officers?" A comparative study between U.S. corporate and nonprofit executive leaders' social media communication strategies. *Nonprofit Management & Leadership, 33*(4), 879–892.

Yuniarti, R., & Ruldeviyani, Y. (2023). Sentiment analysis of banking risk profile determination: The case study of Bank XYZ. *International Conference on Information Technology and Advanced Mechanical and Electronical Engineering, 2865*(1), 1–9.

Zammarchi, G., Mola, F., & Conversano, C. (2023). Using sentiment analysis to evaluate the impact of the COVID-19 outbreak on Italy's country reputation and stock market performance. *Statistical Methods Applications, 32*, 1001–1022.

Zerbini, F. (2017). CSR initiatives as market signals: A Review and research agenda. *Journal of Business Ethics, 146*, 1–23.

Zhang, Y., Chen, F., & Rohe, K. (2022). Social media public opinion as flocks in a murmuration: Conceptualizing and measuring opinion expression on social media. *Journal of Computer-Mediated Communication, 27*(1), 1–22.

Zhang, N., Guo, X., Zhang, L., & He, L. (2021). How to repair public trust effectively: Research on enterprise online public opinion crisis response. *Electronic Commerce Research and Applications, 49*, Article 101077.

Zhang, Y., Shah, D., Foley, J., Abhishek, A., Lukito, J., Suk, J., & Garlough, C. (2019). Whose lives matter? Mass shootings and social media discourses of sympathy and policy, 2012–2014. *Journal of Computer-Mediated Communication, 24*(4), 182–202. https://doi.org/10.1093/jcmc/zmz009

Zhang, N., Wang, X., Guo, X., Cheng, X., & Pang, Z. (2024). How organization response influences public sentiment and behavior in online crisis: The role of response strategies, tenor of discourse and executives. *Public Relations Review, 50*, Article 102395.

Zhao, X., Chen, Y.-R., & R. (2022). How brand-stakeholder dialogue drives brand-hosted community engagement on social media: A mixed-methods approach. *Computers in Human Behavior, 131*, Article 107208.

Zhou, M. (2022). Public opinion on MOOCs: Sentiment and content analyses of Chinese. microblogging data. *Behavior Information Technology, 41*(2), 365–382.

Zhou, Z., & Ki, E. J. (2018). Does severity matter? An investigation of crisis severity from defensive attribution theory perspective. *Public Relations Review, 44*(4), 610–618.

Zhou, Z., Zhou, X., Chen, Y., & Qi, H. (2024). Evolution of online public opinion on major accidents: Implications for post-accident response based on social media network. *Expert Systems with Applications, 235*, Article 121307 (Article).